UDCT v7.14 Preregistration: QUMOND Phase-Space Consistency and Independent Validation Protocol

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GitHub: https://github.com/wspaik5/UDCT_v7.14_Preregistration

STATUS: PREREGISTRATION UPLOAD CANDIDATE. DOI RESERVED, NOT YET PUBLISHED. NO PROTOCOL-DRIVEN CONFIRMATORY TEST EXECUTED.

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Status: FINAL UPLOAD CANDIDATE (DOI RESERVED; NOT PUBLISHED) — NOT YET PUBLISHED OR REGISTERED; NO CONFIRMATORY RUNS EXECUTED UNDER THIS PROTOCOL.

Study type: Prospective, finite-grid, model-conditional numerical protocol; no observational inference.

Abstract

Archived UDCT v7.14 Steps 01–50 and subsequent exploratory audits reported two separate features of a specified isolated spherical simple-QUMOND/Plummer model: (a) cancellation of a leading central asymptotic coefficient at tracer exponent p=1.5; and (b) a finite-energy local Osipkov–Merritt (OM) distribution-function tangency candidate near r_a/a=0.466466787396. Neither is a new prospective discovery. This document specifies a prospective 20-point numerical DF-sign audit, three-valued decision rules, source provenance, and error-control gates. Crucially, the proposed coordinates have not been independently checked against every archived exploratory run for prior exposure. They are a prespecified future grid, not a verified held-out sample. No global DF-positivity theorem, novel gravity law, full GQUMOND solution, or direct Hydrus I stellar-likelihood result is claimed.

1. Retrospective scientific basis

The original v7.14 Step 01–50 archives include Hydrus I-motivated sensitivity experiments, three-dimensional Hessian diagnostics, Jeans second-moment calculations, and later isolated spherical OM inversion. The later branch uses a different mathematical question than the early 3D Hessian tests. Six exploratory audits and subsequent high-precision quadrature comparisons investigate the finite-window tangency. The existing results are retrospective; no reuse of them as prospective confirmatory evidence is permitted. The Phase I integrated audit checks source identity and file integrity but is not an independent physics replication.

2. Fixed physical model and conventions

The gravitating mass is spherical Plummer, with g_N(r)=G M r/(r^2+a^2)^(3/2). In the isolated spherical simple-QUMOND construction, g(r)=g_N(r)/2+sqrt(g_N(r)^2/4+a_0*g_N(r)). Tracer rho_p(r)=rho_0*[1+(r/a)^p]^(-5/p), rho_0=1. OM augmented density rho_Q(r)=[1+(r/r_a)^2]*rho_p(r), beta(r)=r^2/(r^2+r_a^2). Reference constants: G=4.30091e-3 pc (km/s)^2/Msun, M=32730 Msun, a=53.3 pc, a_0=1.2e-10 m/s^2. These are model inputs, not observationally fitted parameter-free predictions. Use Phi(r)-Phi(0) as the outward-increasing potential difference; do not substitute an infinity-zero potential in isolated MOND, where the outer potential grows logarithmically. The dimensionless energy label x=r/a means the spherical turning-point label used by archived inversions, not an observed stellar radius.

3. Pre-specified grid and its limitation

The 20 prospective evaluation points are the Cartesian product p in {1.487, 1.513}, r_a/a in {0.44, 0.48}, and energy label x in {0.30,0.40,0.50,0.60,0.70}. Exact enumeration is in PRESPECIFIED_GRID.csv. This grid is not claimed to be untouched by prior exploration: historical exposure is not fully audited. It is deliberately separated from the previously reported p=1.5 tangency and

serves a prospective reproducibility/stress-test objective rather than a statistically independent discovery claim. No grid points may be changed after publication without a labeled amendment.

4. Analysis and numerical verification plan

Two mathematically equivalent OM inversion formulations should be evaluated, one using the second potential derivative of augmented density with analytic central-endpoint subtraction, and another independently implemented by integration by parts using first derivatives and explicit boundary terms. Both must use the exact same physical model and energy convention. Since these representations share the same underlying inversion kernel, their agreement is a numerical cross-check, not independent physics validation. An implementation using separately written code and independently checked signs/units is required before any certified sign assertion. Preserve the exact Python environment, code hashes, commands, raw integration outputs and diagnostic logs.

Numerical error accounting must cover inner potential quadrature, outer OM quadrature, outer-tail truncation, central endpoint treatment, differentiation/interpolation and floating-point rounding. Prior 55-digit arithmetic and 40/80/160-by-64/128/256 quadrature scans are exploratory diagnostics, not certified enclosures. A validated numerical enclosure must be supplied before labeling a sign certified. If that cannot be established, classify the result as inconclusive. This preregistration does not assert that a validated numerical enclosure implementation already exists.

5. Controls and locked decisions

Control C1: compare the Newtonian Plummer isotropic Jeans radial second moment at r=a against the analytic reference G*M/(6*sqrt(2)*a), with relative discrepancy <= 1e-4. This checks units and integration plumbing, not OM inversion correctness. Control C2: for each evaluation coordinate, compare the two OM inversion representations; their numerical discrepancy must be <= 1e-6 times max(1,abs(f_A),abs(f_B)) in archived normalization. C2 is an operational screening tolerance, not a certified error bound. Control C3: verify that the augmented-density expression reduces to the isotropic tracer as r_a tends to infinity at fixed r. Report all failures. The main sign decision requires controls C1–C3 and justified bounds:

• Certified negative at a sampled point: validated upper bound U<0 and controls pass.

• Certified nonnegative at all sampled points: validated lower bound L>=0 for each of 20 points and controls pass. This does not imply continuum or global positivity.

• Inconclusive: interval overlaps zero, validated bounds unavailable, controls fail, or implementations disagree. Uncertified negative point estimates cannot be called a certified failure.

The primary report must enumerate all 20 points, including inconclusive and failed controls. There is no requirement to force a positive or negative conclusion. Any unplanned scan is exploratory and reported separately.

6. Sequence and publication lock

1. Publish this protocol, PROTOCOL.json, grid CSV, provenance manifest and the source-code status before new protocol-driven tests.

2. Verify the independent inversion derivation and complete a documented numerical error-enclosure implementation; amendments are allowed only if timestamped and identified, never silently retrofitted.

3. Run controls and the 20-point grid; archive every result, error interval and execution log.

4. Release a distinct Post-registration result package, including failures and deviations.

Publication caveat: Because the independent implementation and rigorous enclosure are not yet delivered, the present package preregisters the decision logic and grid but is not an executable certified-sign test. If published as is, describe it as a *prospective protocol with implementation gates*, not as a fully locked validated numerical analysis pipeline.

7. Scientific scope and exclusions

The central p=1.5 coefficient cancellation is a local asymptotic observation and does not establish the finite-energy OM tangency; the latter is a local numerical feature, not a universal critical anisotropy radius. Negative DF excludes only a specified model/tracer/anisotropy ansatz under the stated assumptions, not MOND or QUMOND generally. Hessian eigenvalues, Jeans second moments, DF positivity and dynamical stability are different tests. No 3D external-field solution, full GQUMOND higher-derivative action, or novel modified-gravity law is tested here.

Hydrus I informed aspects of the broader program but the current tests do not fit the complete individual-star line-of-sight velocity data. Such a fit would require a separate preregistration covering membership, selection, foregrounds, binaries, uncertainties and likelihood comparisons.

8. References

1. Bekenstein, J. & Milgrom, M. (1984), Astrophysical Journal 286, 7–14, doi:10.1086/162570.

2. Milgrom, M. (2010), MNRAS 403, 886–895, doi:10.1111/j.1365-2966.2009.16184.x.

3. Milgrom, M. (2023), Physical Review D 108, 084005, doi:10.1103/PhysRevD.108.084005.

4. Merritt, D. (1985), MNRAS 214, 25P–28P.

5. Binney, J. & Tremaine, S. (2008), Galactic Dynamics, 2nd ed.

6. Koposov, S. E. et al. (2018), MNRAS 479, 5343–5361, doi:10.1093/mnras/sty1772.

7. Paik, W. S. (2026), UDCT v7.14 Exploratory QUMOND Hessian and Phase-Space Consistency Tests (Steps 01–50), archived original materials.

8. Paik, W. S. (2026), UDCT v7.14 Exploratory Findings: The p=1.5 Boundary in QUMOND, archived audits 01–06.

Extended Research Record and Archival Appendix

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RETROSPECTIVE ARCHIVE. All entries below predate this preregistration. Their presence and checksum do not establish independently rerun or scientifically validated calculations.

A. Research provenance and evidence boundaries

The original Steps 01–50 encompass Hydrus I-motivated sensitivity, 3D field/Hessian diagnostics, spherical Jeans second moments, and isolated spherical distribution-function inversion. These are distinct branches with different mathematical claims.

The six exploratory audits describe the p=1.5 central asymptotic leading-term cancellation and a model-specific finite-energy Osipkov–Merritt local tangency near r_a/a=0.466466787396. The two features are not the same boundary.

The subsequent 46-item package records further numerical investigations, including quadrature order, central endpoint subtraction, tail treatment, derivative sensitivity, high precision, and preregistration readiness. File titles identify the audit topics, not independently verified outcomes.

Neither archival reproduction nor an agreement between two integral forms proves global DF positivity, full GQUMOND field solutions, dynamical stability, or direct Hydrus I individual-star likelihood agreement.

The prospective 20-point protocol is separate from this historical archive. Historical non-exposure of the planned grid has not been established. Certified sign classification requires justified numerical enclosures; the provided Python script validates protocol metadata only.

B. Preserved original source archives

B.1 01_Original_Exploratory_Steps_01-25.zip

Original source: 7.14 Hessian 1~25.zip

SHA-256: 24e646cb99f0b0cd0f997e9afba68f2be3d3a01f68ad7b222e3d943cb66924ff

Preserved without altering archive contents.

B.2 02_Original_Exploratory_Steps_26-50.zip

Original source: 7.14 Hessian 26~50.zip

SHA-256: 14da6463e2dc1b9af6726c448517aa6013709ae2e967a3b5cb8d7ae985292f58

Preserved without altering archive contents.

B.3 03_Original_Exploratory_Audits_01-06.zip

Original source: 7.14 Exploratory audit 1~6.zip

SHA-256: 6980b333823abba340403d6d921b50805291e42b5f8b810ab003bce19ef44614

Preserved without altering archive contents.

B.4 04_Followup_Audit_Records_46_Items.zip

Original source: 7.14g ■ ■■ ■■■■ ■■ ■■ 1~46 (pdf).zip

SHA-256: b94e54f045542adfcefa05f6f71198370218e9c2e616e98f243c3dfd5f46fa25

Preserved without altering archive contents.

B.5 05_Original_Reproducibility_Bundle.zip

Original source: UDCT_v7.14_Reproducibility_Download.zip

SHA-256: bfbbd88ab4ce6e79b46c2d13eaf438ce6bb789ea50590ac5250b675e40e8bae1

Preserved without altering archive contents.

B.6 06_p15_Reproducibility_Bundle.zip

Original source: UDCT_v714_p15_Reproducibilty package.zip

SHA-256: 94d298b74fb70b8780d4b8d02a102a0f03fc6efddeefcdf027851f4d13098059

Preserved without altering archive contents.

C. Follow-up numerical record: all 46 uploaded entries

C.01 UDCT_v714_Central_Asymptotic_Subtraction_REISSUED_20261110_v2_20261110_125051_841.zip

Archived size: 5482 bytes. Type: .zip. Nested entries: 8. Nested ZIP integrity: pass.

`SHA-256: f08d9bd738b1c2c3cd28fa5c99b2438e9751123b50e6b218908b64c7f0c5a803`

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.02 UDCT_v714_Central_Convergence_Audit_20261110_20261110_125043_851.zip

Archived size: 4409 bytes. Type: .zip. Nested entries: 7. Nested ZIP integrity: pass.

`SHA-256: 51386f2b280587df8929f4a2c93ddce646b99ef1bef85acbc67c369351a39056`

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.03 UDCT_v714_Central_Regularity_Gate_REISSUED_20261110_v2_20261110_125028_970.zip

Archived size: 8013 bytes. Type: .zip. Nested entries: 10. Nested ZIP integrity: pass.

`SHA-256: f927d13d974186d5943b88f652e19a157d4eff167d6c7a6ff69041aa84180379`

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.04 UDCT_v714_Central_Subtraction_Convergence_REISSUED_20261110_v2_20261110_125057_940.zip

Archived size: 6424 bytes. Type: .zip. Nested entries: 7. Nested ZIP integrity: pass.

`SHA-256: 4d96a1e91adf0a7d64353596500d6ce0abdc7d4b333a89d7a8ca8de6beb686b1`

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.05 UDCT_v714_Convergence_Audit_FRESH_20261031_v2_20261110_124829_235.zip

Archived size: 5248 bytes. Type: .zip. Nested entries: 7. Nested ZIP integrity: pass.

`SHA-256: bdf559b5bdfea7a7e6bfb1ce2023825d78c55a248b2b5131d4012a058bf6ed5f`

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.06 UDCT_v714_Extended_Energy_Scan_REISSUED_20261030_20261110_124801_469.zip

Archived size: 5358 bytes. Type: .zip. Nested entries: 6. Nested ZIP integrity: pass.

`SHA-256: e836c2e80ac142473a0920c8bf235c66bd5f110283c71e487942b6896dd9c672`

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.07 UDCT_v714_Extended_OM_Scan_20261109_20261110_124933_562.zip

Archived size: 6772 bytes. Type: .zip. Nested entries: 7. Nested ZIP integrity: pass.

`SHA-256: 6f258d9e9277f0114b350a8ee7fbe731ff33813b028f50ec0e31335309188b58`

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.08 UDCT_v714_FD_Step_Convergence_REISSUED_20261110_v2_20261110_125111_151.zip

Archived size: 7666 bytes. Type: .zip. Nested entries: 9. Nested ZIP integrity: pass.

SHA-256: f5a85b21d8eced2234897166784aa78f31b385b929fb1d6a464ce34a6f6f91ef

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.09 UDCT_v714_Finite_Minima_REISSUED_20261031_v2_20261110_124836_178.zip

Archived size: 6095 bytes. Type: .zip. Nested entries: 7. Nested ZIP integrity: pass.

SHA-256: 6b2759813aba6f35a9a7250870eef76650e24d0830bc5d53c86bc8c92e1bb42a

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.10 UDCT_v714_Finite_Outer_Tail_Audit_REISSUED_20261110_v2_20261110_125018_596.zip

Archived size: 6961 bytes. Type: .zip. Nested entries: 8. Nested ZIP integrity: pass.

SHA-256: 31ef6026d7efcb469d377d11aff3ddbb74213e1915cc38c82a24a91cdb43164e

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.11 UDCT_v714_Fresh_Local_Rerun_20261030_20261110_124740_169.zip

Archived size: 8176 bytes. Type: .zip. Nested entries: 11. Nested ZIP integrity: pass.

SHA-256: d8d48fcee65a0e4c312eec4bf037d7a187a16948389eb359772bcb5d84abc4ef

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.12 UDCT_v714_High_Precision_OM_Crosscheck_REISSUED_20261110_v2_20261110_125130_153.zip

Archived size: 6068 bytes. Type: .zip. Nested entries: 8. Nested ZIP integrity: pass.

SHA-256: 25c3008b8f4ac327ededb868a37b52ef81bd932b58a4ea6162db7c796a3cc354

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.13 UDCT_v714_Independent_OM_20261109_20261110_124913_839.zip

Archived size: 8615 bytes. Type: .zip. Nested entries: 9. Nested ZIP integrity: pass.

SHA-256: 73842527d6331ce5baef1803dbf67051f4cdc94cca109df53a7a5403146d3438

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.14 UDCT_v714_Independent_OM_Crosscheck_FRESH_v2_20261110_124843_115.zip

Archived size: 4859 bytes. Type: .zip. Nested entries: 6. Nested ZIP integrity: pass.

SHA-256: 35896e91ed9909c57750a6f5072db368777be2c62f37b6fb3d0e9e7112ac3892

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.15 UDCT_v714_Independent_Tangency_Audit_20261109_20261110_124925_712.zip

Archived size: 12305 bytes. Type: .zip. Nested entries: 12. Nested ZIP integrity: pass.

SHA-256: c1b1c5880b498b5aa24a5866072e127ac0d33a84da4f620d8aa1934e140b5a5f

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.16 UDCT_v714_Model_Convention_Gate_REISSUED_20261110_v2_20261110_124959_206.zip

Archived size: 11346 bytes. Type: .zip. Nested entries: 9. Nested ZIP integrity: pass.

SHA-256: 1c648c607d7d132070438e3bf055686bfe16e54de4913764e2c208e84a5834e3

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.17 UDCT_v714_Nested_Quadrature_Gate_REPRO_20261110_20261110_125137_017.zip

Archived size: 7788 bytes. Type: .zip. Nested entries: 9. Nested ZIP integrity: pass.

SHA-256: dce2eee8c3bb772e7c89920cf8e8fe2a526058a6e97c5fabb3cf9038a4853465

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.18 UDCT_v714_Newtonian_OM_Control_REISSUED_20261029_20261110_124620_851.zip

Archived size: 5313 bytes. Type: .zip. Nested entries: 5. Nested ZIP integrity: pass.

SHA-256: 39a150b4ac37db0ac882a4abadacd68d6ae6e51260e45339153c31de4c41038f

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.19 UDCT_v714_OM_Alternative_Integral_Audit_20261110_20261110_125123_536.zip

Archived size: 8052 bytes. Type: .zip. Nested entries: 9. Nested ZIP integrity: pass.

SHA-256: 6fb8c5b772fd59683e1651a9bdbe2089b2342d465a4140fdab6f7bcb44e43c61

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.20 UDCT_v714_OM_Derivative_Crosscheck_REISSUED_20261110_v2_20261110_125104_335.zip

Archived size: 7024 bytes. Type: .zip. Nested entries: 9. Nested ZIP integrity: pass.

SHA-256: 6aa980f9f9f713e37b84d496c1eaf57aaddb3bf0bf53107d90bf06bafe51be7b

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.21 UDCT_v714_OM_Gate_Download_Fix_02_20261110_124731_146.zip

Archived size: 4186 bytes. Type: .zip. Nested entries: 4. Nested ZIP integrity: pass.

SHA-256: bdad29e59af54f5de6a0f2014ed5013730cd1e01bebd894135020aba27c5c7dd

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.22 UDCT_v714_OM_Inversion_REISSUED_20261029_v2_20261110_124537_978.zip

Archived size: 15014 bytes. Type: .zip. Nested entries: 4. Nested ZIP integrity: pass.

SHA-256: a5621db977cc63388b0cabce19607ccae4573ee800ac3f58de96ea3a45fb0855

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.23 UDCT_v714_OM_Outer_Boundary_Audit_REISSUED_20261110_v2_20261110_125005_485.zip

Archived size: 6753 bytes. Type: .zip. Nested entries: 8. Nested ZIP integrity: pass.

SHA-256: fe538452418300d5b81d9c2bdf5a753b6e2de4a66d812e6088263a56d68f602e

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.24 UDCT_v714_Positive_Energy_Window_Gate_REISSUED_20261110_v2_20261110_125036_708.zip

Archived size: 7292 bytes. Type: .zip. Nested entries: 8. Nested ZIP integrity: pass.

`SHA-256: 6b0f580918a3d4d8e019401ea35bab529ecabf1e4449d4c57f4ca44849db13f2`

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.25 UDCT_v714_Precision_OM_Audit_NEW_LINK_20261030_20261110_124813_703.zip

Archived size: 9414 bytes. Type: .zip. Nested entries: 9. Nested ZIP integrity: pass.

`SHA-256: f01e720065ff5be432cedccd7aa2d8f7b0f5c0578081a13a4b8672de81dc6113`

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.26 UDCT_v714_Prospective_Protocol_Gate_DRAFT_20261109_20261110_124951_662.zip

Archived size: 4659 bytes. Type: .zip. Nested entries: 7. Nested ZIP integrity: pass.

`SHA-256: d5fd940ec150db78f245d85d58fdcf091587215298ec3499873a3aa5dbe9e23c`

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.27 UDCT_v714_Protocol_Lock_Readiness_20261110_20261110_125142_622.zip

Archived size: 4237 bytes. Type: .zip. Nested entries: 6. Nested ZIP integrity: pass.

`SHA-256: 2eaeacf0a8ae4e68b0bbf1ef3b656232cc2815ca11e746bf9ef6a0bdd1ef347c`

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.28 UDCT_v714_Protocol01_Asymptotic_Notes_20261110_124440_189.md

Archived size: 2146 bytes. Type: .md. Nested entries: not applicable. Nested ZIP integrity: not applicable.

`SHA-256: 192e5d6be742c3b47a14a9755ad878423d5a65fe6b0072153e8681ea0d472470`

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.29 UDCT_v714_Protocol01_Continuation_20261029_20261110_124643_378.zip

Archived size: 19759 bytes. Type: .zip. Nested entries: 4. Nested ZIP integrity: pass.

`SHA-256: d7f3d7c96f364cade60c25d53bd4f3e4329b70cd1f625d48a23009495bbcd0f5`

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.30 UDCT_v714_Protocol01_CrossQuadrature_20261029_20261110_124702_757.zip

Archived size: 10097 bytes. Type: .zip. Nested entries: 8. Nested ZIP integrity: pass.

`SHA-256: 253a9e521d9cef2a3a1b802f545e181d536d817ec8e6d7a9b7cba0deb2d9f3ac`

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.31 UDCT_v714_Protocol01_DRAFT_20261110_124425_770.json

Archived size: 1258 bytes. Type: .json. Nested entries: not applicable. Nested ZIP integrity: not applicable.

`SHA-256: 61d3a8af5ab98dedfe3cfa9f362f822a3a617fa4c652814cfd8965f6d3492368`

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.32 UDCT_v714_Protocol01_Gravity_Gate_20261029_REISSUED_20261110_124724_009.zip

Archived size: 5030 bytes. Type: .zip. Nested entries: 6. Nested ZIP integrity: pass.

SHA-256: 6eaed323b619dec9ea8ad20681fd397e9a724169a739e9e1da7906b06c94cb08

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.33 UDCT_v714_Protocol01_Independent_Asymptotic_20261110_124442_637.py

Archived size: 1190 bytes. Type: .py. Nested entries: not applicable. Nested ZIP integrity: not applicable.

SHA-256: 3d62cf7b512cbe5b44eab73a369d8c86e6f260f5e2de0ccb30a73b38b6e0d31f

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.34 UDCT_v714_Protocol01_NextStage_DesignGate_20261029_20261110_124515_076.zip

Archived size: 10553 bytes. Type: .zip. Nested entries: 3. Nested ZIP integrity: pass.

SHA-256: 79324efed04d9c23a0d51a8f4fd0ac4c9aece335abfb2cfe1162bfe8fd9bd69f

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.35 UDCT_v714_Protocol01_NextStage_DesignGate_20261029_20261110_124522_601.zip

Archived size: 10553 bytes. Type: .zip. Nested entries: 3. Nested ZIP integrity: pass.

SHA-256: 79324efed04d9c23a0d51a8f4fd0ac4c9aece335abfb2cfe1162bfe8fd9bd69f

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.36 UDCT_v714_Protocol01_OM_NecessaryGate_20261029_20261110_124531_515.zip

Archived size: 3796 bytes. Type: .zip. Nested entries: 5. Nested ZIP integrity: pass.

SHA-256: 6e3b7e43547ff358334000bc5c4e58d94a6c442543a7debfe09d5bc48ab0d51a

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.37 UDCT_v714_Protocol01_Prereg_Candidate_20261109_20261110_124944_011.zip

Archived size: 6162 bytes. Type: .zip. Nested entries: 6. Nested ZIP integrity: pass.

SHA-256: 13c8136a13630672885293435e2c89b97d897e6d8bafb3a6445fc8cb14cf30ab

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.38 UDCT_v714_Protocol01_Preregistration_DRAFT_20261110_124428_397.md

Archived size: 4336 bytes. Type: .md. Nested entries: not applicable. Nested ZIP integrity: not applicable.

SHA-256: aa2e6ed3b5cd8fd3ef9131740d5c28fb5cc1b1ff724e6a27a9575568e0169613

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.39 UDCT_v714_Protocol01_Reproducibility_20261110_124610_259.zip

Archived size: 3038 bytes. Type: .zip. Nested entries: 5. Nested ZIP integrity: pass.

SHA-256: c6f1c48175175947563d32b9b4e6dcf8480d89dd24e8f4960959514d043c5d4b

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.40 UDCT_v714_Protocol01_Reproducibility_REISSUED_20261029_20261110_124622_899.zip

Archived size: 3038 bytes. Type: .zip. Nested entries: 5. Nested ZIP integrity: pass.

SHA-256: c6f1c48175175947563d32b9b4e6dcf8480d89dd24e8f4960959514d043c5d4b

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.41 UDCT_v714_Protocol01_Verification_REISSUED_20261029_20261110_124508_583.zip

Archived size: 7961 bytes. Type: .zip. Nested entries: 10. Nested ZIP integrity: pass.

SHA-256: e929e3aa0577b1cbd9276ba5656c8cce8d4d87b32edcc1f6dcfdf8ca42f1a079

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.42 UDCT_v714_Sensitivity_Audit_Fresh_Link_20261029_20261110_124715_183.zip

Archived size: 7694 bytes. Type: .zip. Nested entries: 8. Nested ZIP integrity: pass.

SHA-256: 44465f2e0c22492f5102d2621cf6d0a2fa29e876de7204ee0679c2518c38e82e

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.43 UDCT_v714_source_inventory_20261110_124418_886.csv

Archived size: 18818 bytes. Type: .csv. Nested entries: not applicable. Nested ZIP integrity: not applicable.

SHA-256: 9f8b166baf11ff178b59ed36d6e64fd2bb4a88f08b07acc32b409a477ce86d7a

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.44 UDCT_v714_Step50_Kernel_Audit_20261109_20261110_124905_847.zip

Archived size: 10698 bytes. Type: .zip. Nested entries: 9. Nested ZIP integrity: pass.

SHA-256: 7d923eff9c5ec71650974ac944bb9b8dcd24001701e9af20d989130ae93bea89

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.45 UDCT_v714_Step50_Source_Recovery_20261109_20261110_124858_579.zip

Archived size: 9922 bytes. Type: .zip. Nested entries: 11. Nested ZIP integrity: pass.

SHA-256: f169a3632e97eb4ec52d3e2b4bb5fea8f57337be7d33bd75f9fe6a0592642c5b

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

C.46 UDCT_v714_Tail_Adaptive_Crosscheck_REISSUED_20261030_20261110_124821_346.zip

Archived size: 4677 bytes. Type: .zip. Nested entries: 5. Nested ZIP integrity: pass.

SHA-256: 5266e30d936033360808ad77fc4cf90d1300a94815ce17edc95809636e1f5532

Evidence status: archived retrospective artifact. This index does not independently verify its numerical conclusions. See the unchanged source file for inputs, outputs and method details.

D. Preregistration boundary and limitations

This volume is archival evidence, not prospective confirmatory data.

The preregistration primary grid contains 20 prespecified coordinates. Historical exposure has not been exhaustively excluded.

Numerical signs are inconclusive unless validated enclosures and controls support them.

Negative DF in a specified ansatz is not a general falsification of MOND or QUMOND.

Hydrus I-motivated model scales are not a fit to the complete observed individual-star velocity sample.

All source archives are separately included in the publication package; they are not embedded in the PDF.

UDCT v7.14: Exploratory QUMOND Hessian and Phase-Space Consistency Tests (Steps 01–50)

Comprehensive research paper and source-by-source technical commentary | First public version | 26 October 2026

Won Shik Paik | Independent Researcher, Auckland, New Zealand

Reserved DOI: https://doi.org/10.5281/zenodo.23663161

GitHub: https://github.com/wspaik5/UDCT_v7.14_PrePreregistration_Exploratory

Status: Pre-preregistration exploratory record. Neither peer reviewed nor a direct observational validation. The DOI remains reserved until Zenodo publication.

Abstract

This extended manuscript documents a 50-stage computational program progressing from Hydrus I-motivated observational sensitivity checks to three-dimensional QUMOND field/Hessian audits, Jeans-equation consistency tests, and isolated spherical distribution-function (DF) inversions. Its purpose is to preserve assumptions, numerical artifacts, mathematical arguments, test boundaries, and failure modes rather than announce a new gravitational law. The later branch employs a Plummer gravitating source and the tracer family rho*(r)=rho0[1+(r/a)^p]^(-5/p). Under specific central scaling and Abel-inversion regularity conditions, the leading asymptotic DF coefficient changes sign across p=3/2; at exact equality it cancels and cannot establish the full DF sign. Archived Osipkov–Merritt (OM) calculations report a finite-window local tangency r_a/a=0.46646678739601644 for p=1.5, with energy-label r/a=0.47409755283222904. Neither a finite-window root nor a negative DF in this ansatz is a global MOND falsification. The early 3D Hessian computations are distinct from the spherical DF branch and do not by themselves implement the generalized GQUMOND field equations. The companion archive preserves original source code, CSV/JSON records and per-step instructions. No full 50-stage independent replication or individual-star Hydrus I likelihood fit is claimed.

Reader guide and central conclusions

• The first part presents physical motivation, equations, inversion assumptions, interpretation and limits; the second part provides the 50-step provenance narrative and detailed file-level audit.

• The Hessian, Jeans second moment, phase-space DF, and dynamical stability are separate mathematical tests; no inference between them is automatic.

• The local p=1.5 coefficient cancellation is a particularly important unresolved equality case, not a proven global threshold.

• The Step 50 anisotropy value is local to the declared scan window; all-energy admissibility requires further tests.

• Figures are redrawn from archived CSV outputs. This document does not represent a new independent computational campaign.

1. Scientific lineage and research question

Bekenstein and Milgrom (1984) introduced AQUAL as a nonlinear modified-gravity theory. Milgrom (2010) formulated QUMOND as a quasilinear construction. Milgrom (2023) developed GQUMOND actions depending on higher spatial derivatives of the Newtonian potential. This 2023 article is crucial prior art for Hessian-sensitive modifications; a diagnostic evaluation of a Hessian tensor is not equivalent to deriving or solving the full action-dependent GQUMOND field equations. The current 3D Hessian branch is described as a diagnostic and numerical validation exercise, while the later spherical DF branch is tied to a restricted QUMOND potential.

The research question is not which gravity camp wins. It is whether specified combinations of gravitational potential, stellar tracer density and orbital anisotropy can be represented by a physically admissible, nonnegative equilibrium distribution function. A failure excludes that particular model/ansatz under the stated assumptions, not every model in its parent theory.

2. Data provenance and the two research branches

The supplied primary sources are 7.14 Hessian 1~25.zip (80 top-level entries) and 7.14 Hessian 26~50.zip (25 nested step packages). The first archive contains a mixture of early experimental scripts, data tables and numbered later packages; some early files cannot be uniquely assigned to Steps 01–15. The second archive has explicit Steps 26–50. Source-file names are kept rather than retrospectively inventing a complete chronology. ZIP integrity, ability to execute Python, numerical reproducibility, correct equations and observational agreement are different levels of evidence.

Early work explored membership, foreground, metallicity, velocity gradients, uncertainty, baseline Newtonian comparisons and other Hydrus I-motivated issues. Steps 16–25 focused on 3D fields and Hessian convergence, boundaries and rotations. Steps 26–38 transitioned through Jeans and DF inversion. Steps 39–44 examined critical exponents, independent quadrature, forward reconstruction and tail sensitivity. Steps 45–50 explored OM anisotropy, sign maps, zero contours, lobe collapse and local tangency.

3. Field models and the 3D Hessian

For a Newtonian potential Phi_N, the Hessian is H_ij = partial_i partial_j Phi_N. Under an orthogonal rotation R, the exact tensor transforms H -> R H R^T, preserving trace and eigenvalues. Discrete Cartesian finite differences need not preserve this identity exactly; convergence and rotation audits quantify numerical artifacts.
Boundary-condition effects, finite-box size and external-field surrogates must be distinguished from genuinely covariant field equations. Eigenvalue signs can indicate local tidal geometry, but do not establish the sign of a stellar distribution function.

In conventional QUMOND, a Newtonian Poisson solve supplies the Newtonian field, and a second Poisson solve obtains the modified potential using a nonlinear source built from the Newtonian acceleration. GQUMOND permits additional higher-derivative dependence in its action, changing the effective source. One cannot infer implementation of a GQUMOND Lagrangian merely from Hessian arrays being computed. The current release retains the distinction as a reproducibility and prior-art safeguard.

4. Spherical source, tracer and energy variables

The late DF branch assumes an isolated spherical Plummer-type gravitating source and a separately specified tracer family rho_star(r) = rho0 [1+(r/a)^p]^(-5/p). The source mass density and tracer density need not coincide. The source length scale quoted in the earlier manuscript is a=53.3 pc, which is an archived modeling choice rather than a universal galaxy constant. The potential convention matters: DeltaPhi=Phi(r)-Phi(0) increases outward, whereas some standard Eddington presentations use the relative potential Psi=-Phi+Phi(infinity). Equations must not mix these conventions without changing integration limits and boundary terms.

For a cored spherical Newtonian source, the inner Newtonian acceleration scales as g_N~r. In the assumed deep-MOND central regime, g~sqrt(a0*g_N)~r^(1/2). Therefore DeltaPhi~r^(3/2). The tracer expansion at small r is rho_star(r)=rho0[1-(5/p)(r/a)^p+O(r^(2p))]. Eliminating r gives rho(DeltaPhi)=rho0-D(DeltaPhi)^s+..., with D>0 and s=2p/3. Each proportionality assumes the stated central regime and smoothness; an external field or another central mass law can change the exponent.

5. Eddington/Abel inversion and positivity

An isotropic equilibrium DF f(E) produces a density by integration over allowed velocities. With absolute potential and E=v^2/2+Phi(r), rho(Phi)=4*pi*sqrt(2)*Integral_{Phi}^{infinity} f(E)*sqrt(E-Phi) dE for the appropriate convention. Inverting this Abel-type transform involves differentiating the density as a function of potential and accounting for endpoints. For the conventional relative-potential convention, f(Eps)=(1/(sqrt(8)*pi^2))[Integral_0^Eps dPsi (d^2rho/$dPsi^2$)/sqrt(Eps-Psi)+(1/sqrt(Eps))(drho/dPsi)|Psi=0]. The exact signs and boundaries depend on which energy convention is chosen.

A negative DF at an accessible phase-space point contradicts a nonnegative distribution for the specified equilibrium density/potential pair and chosen inversion. It is not necessarily a negative local velocity dispersion, which is a second moment of a distribution, nor proof of linear instability. Numerical inversion can also produce false negative lobes if energy sampling, truncation, interpolation, endpoint subtraction or derivative regularization is inaccurate; independent quadrature and forward reconstruction are therefore essential controls.

6. Central asymptotics and the critical exponent

Let s=2p/3 and rho(DeltaPhi)=rho0-D(DeltaPhi)^s+..., D>0. The archived derivation records a leading singular contribution proportional to -D*s*(s-1)*Beta(1/2,3/2-s)*E^(s-3/2), subject to convergence and dominance assumptions. For 0<s<3/2 the beta factor is positive. Accordingly the displayed coefficient is positive when 0<p<1.5 and negative when 1.5<p<2.25. These statements concern the leading singular term in the stated range, not every energy of the complete inverted DF.

At p=1.5, s=1 and the factor (s-1) vanishes. This does NOT imply f=0, positivity, or negativity: the next nonzero term, regular contributions, and any boundary term can dominate. Near p=2.25 the beta-function expression reaches a convergence boundary; extension outside the stated interval requires separate asymptotics. A rigorous equality-case conclusion requires an independent derivation of the next-order potential and tracer expansions, inversion of those terms, and a numerical check at decreasing energy scales.

7. Osipkov–Merritt orbital anisotropy

For the standard radially anisotropic OM family, beta(r)=r^2/(r^2+r_a^2), and the augmented tracer density is rho_aug(r)=(1+r^2/r_a^2)rho_star(r). The DF depends on Q=Eps-L^2/(2r_a^2), subject to the selected potential convention. Because the Abel inversion is linear in augmented density, f(Q;r_a)=f_iso(Q)+B(Q)/(r_a/a)^2 for a fixed potential and tracer. This decomposition is useful for verifying contour roots and numerical monotonicity, but the positivity domain remains a function of all accessible Q. A tangency of a sampled negative lobe is not a proof of a global critical anisotropy radius.

8. Archived finite-window sign boundaries

The Step 48 example p=1.5, r_a/a=0.4 reports two zero labels r/a≈0.22370244 and 0.76696971 and a normalized minimum DF ≈-0.0008538448. These r/a values are energy labels induced by the monotonic spherical potential, not literal boundaries of a spatial shell of unphysical stars. Step 50 scans the specified window 0.25≤r/a≤0.75 and reports local tangency r_a/a=0.46646678739601644 at label r/a=0.47409755283222904. Nearby archived results: p=1.49 -> r_a/a=0.462739644607, label r/a=0.476369715642; p=1.501 -> r_a/a=0.466841165090, label r/a=0.473866183835.

The finite-difference derivative near the reported tangency is close to zero, but a small residual is a local numerical diagnostic. Comparisons of quadrature orders test internal numerical stability; they do not independently establish that the potential, tracer, DF kernel or model assumptions are correct.

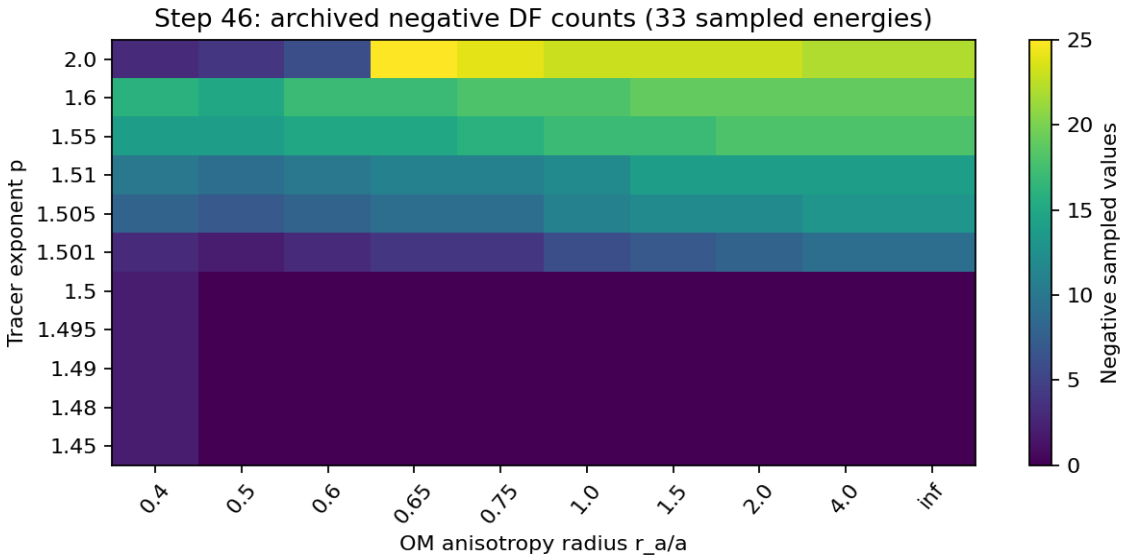


Figure 1. Step 46 archived discrete negative-DF count map; not an all-energy positivity certificate.

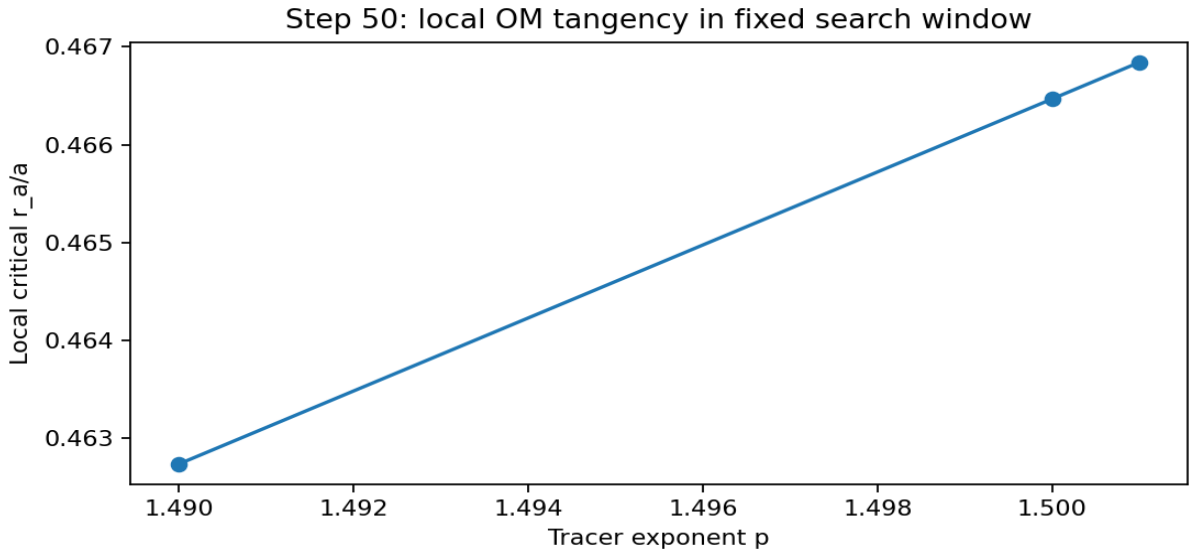


Figure 2. Step 50 local tangency values for nearby tracer exponents.

9. Jeans equations, Hessian, DF and stability: non-equivalent tests

The spherical Jeans equation constrains the tracer-weighted radial second moment: d(rho*sigma_r²)/dr + 2*beta*rho*sigma_r²/r = -rho*dPhi/dr. Solutions depend on boundary conditions, density and anisotropy; a negative formal second moment is a different inconsistency from a negative inverted DF. A Hessian eigenvalue describes curvature of the gravitational potential. A nonnegative DF is a stronger equilibrium phase-space requirement under a chosen ansatz. Dynamical stability requires additional perturbative or nonlinear time-evolution analysis. No one of these four signs can be substituted for another.

10. Observational motivation versus direct Hydrus I inference

Hydrus I observational work motivates scales and dwarf-galaxy stress tests, but using observation-informed model inputs is not equivalent to fitting the full individual-star line-of-sight velocity catalogue. A direct analysis needs per-star velocities, measurement errors, membership probabilities, foreground distributions, binary-star contamination, selection functions, distance and systemic velocity nuisance parameters, and likelihood comparisons across gravitational models. The present late spherical DF calculations are not a star-by-star fit. They are not a full 3D external-field DF inversion either.

11. Numerical credibility and audit tiers

Tier 1 — provenance: original ZIP, code and output files retained byte-for-byte, with checksums.

Tier 2 — execution: source script runs in a documented environment without failure.

Tier 3 — repeatability: regenerated CSV/JSON agree within stated tolerances or byte-for-byte.

Tier 4 — independent numerical method: distinct integration, interpolation and endpoint treatment agree.

Tier 5 — mathematical verification: action, field equation, potential and inversion are derived consistently.

Tier 6 — empirical validation: held-out stellar data are fitted under preregistered criteria and compared with alternatives.

Earlier internal records reported selected reexecution of five scripts (Steps 39, 42, 46, 47 and 50) and matching 13 CSV/JSON outputs. The current expanded manuscript audits archived source content and includes original files; it does not newly establish independent execution of all 50 steps. Reproducing a result with the same implementation does not guarantee independent mathematical correctness.

12. Detailed archived source inventory: early Steps 01–25

The following list preserves filenames exactly as supplied. It is not a verified one-to-one Step 01–25 chronology. A few entries may be numbered subpackages; early scripts are otherwise grouped by topic. Original files remain in the bundled ZIP.

audit_hydrus_20261026_021810_002.py | audit_hydrus_20261026_021854_287.py | bootstrap_replicates_20261026_021942_229.csv | bootstrap_stability_20261026_021944_811.py | bootstrap_summary_20261026_021947_149.json | center_anisotropy_20261026_022038_265.py | comparison_20261026_021957_126.csv | comparison_20261026_022211_162.csv

data_audit_20261026_021847_544.json | data_audit_20261026_021849_724.csv | efe_sensitivity_20261026_022224_937.csv | efe_surrogate_20261026_022226_922.py | exploratory_fit_20261026_021904_415.json | exploratory_fit_20261026_021906_264.py | fixed_mass_profile_20261026_022053_021.csv | gradient_sensitivity_20261026_021911_953.json

gradient_sensitivity_20261026_021913_880.py | J_MNRAS_479_5343_20261026_021817_471.mht | J_MNRAS_479_5343_20261026_021825_570.mht | J_MNRAS_479_5343_20261026_021900_639.mht | joint_membership_20261026_022123_743.csv | joint_velocity_metallicity_20261026_022126_149.py | membership_sensitivity_20261026_022137_636.csv | mixture_fit_20261026_021928_530.py

mixture_membership_diagnostics_20261026_021930_750.csv | mixture_results_20261026_021933_288.json | newtonian_jeans_baseline_20261026_022012_283.py | newtonian_jeans_results_20261026_022014_583.json | om_comparison_20261026_022151_944.csv | om_df_test_20261026_022155_524.py | README_20261026_021814_560.md | README_20261026_021851_500.md

README_20261026_022016_760.md | README_20261026_022024_698.md | README_20261026_022040_411.md | README_20261026_022140_172.md | results_20261026_022042_528.csv | results_20261026_022128_195.json | scale_distance_results_20261026_022026_708.csv | scale_distance_sensitivity_20261026_022028_496.py

sensitivity_20261026_022001_190.py | spatial_sensitivity_20261026_022142_414.csv | spherical_mond_jeans_20261026_022213_884.py | stellar_mass_comparison_20261026_022107_024.py | stellar_mass_diagnostic_20261026_022055_176.py | stellar_only_grid_20261026_022110_321.csv | summary_20261026_022003_389.json | summary_20261026_022044_255.json

summary_20261026_022058_322.json | summary_20261026_022112_149.json | summary_20261026_022159_098.json | summary_20261026_022216_139.json | summary_20261026_022230_319.json | table2_20261026_021828_693.txt |

table2_20261026_021903_065.txt | UDCT_v7_14_Step16_REDOWNLOAD_20261024_20261026_022240_494.zip

UDCT_v7_14_Step17_Boundary_Hessian_Audit_20261026_022248_250.zip |
UDCT_v7_14_Step18_Analytic_Hessian_Benchmark_20261026_022254_170.zip |
UDCT_v7_14_Step19_Convergence_Audit_20261026_022302_780.zip |
UDCT_v7_14_Step20_High_Resolution_Hessian_Audit_20261026_022308_779.zip |
UDCT_v7_14_Step21_Convergence_Reaudit_20261026_022315_272.zip |
UDCT_v7_14_Step22_External_Field_Boundary_20261024_20261026_022326_711.zip |
UDCT_v7_14_Step22_External_Field_Boundary_20261024_20261026_022507_212.zip |
UDCT_v7_14_Step23_3D_Rotation_Covariance_20261024_20261026_022524_581.zip

UDCT_v7_14_Step24_Oblique_Rotation_20261024_20261026_022530_355.zip |
UDCT_v7_14_Step25_3D_Oblique_Hessian_20261024_20261026_022537_848.zip |
UDCT_v7.14_Bootstrap_Stability_Exploratory_20261026_021949_596.zip |
UDCT_v7.14_Center_Anisotropy_Exploratory_20261026_022046_326.zip |
UDCT_v7.14_Exploratory_Mixture_Audit_20261026_021935_461.zip |
UDCT_v7.14_Fixed_Mass_Profile_Exploratory_20261026_022100_563.zip |
UDCT_v7.14_Foreground_Error_Sensitivity_Exploratory_20261026_022005_783.zip |
UDCT_v7.14_Gradient_Exploratory_20261026_021917_062.zip

UDCT_v7.14_Joint_Chemo_Kinematic_Exploratory_20261026_022130_572.zip |
UDCT_v7.14_Newtonian_Jeans_Exploratory_20261026_022019_017.zip |
UDCT_v7.14_Newtonian_Scale_Distance_Sensitivity_20261026_022030_550.zip |
UDCT_v7.14_Physical_Anisotropy_DF_Exploratory_20261026_022202_493.zip |
UDCT_v7.14_Scalar_EFE_Sensitivity_Exploratory_20261026_022233_061.zip |
UDCT_v7.14_Spatial_Membership_Sensitivity_20261026_022145_917.zip |
UDCT_v7.14_Spherical_MOND_Newtonian_Exploratory_20261026_022218_458.zip |
UDCT_v7.14_Stellar_Luminosity_Newtonian_Exploratory_20261026_022114_139.zip

13. Detailed stage-by-stage technical archive: Steps 26–50

For each numbered package below, the section records the source files and excerpts of the archived README or summary. Excerpts are evidence of the author-supplied research record, not independent certification. Original README, Python and machine-readable results are available in the accompanying ZIP. Long README files are excerpted rather than silently rewritten.

Step 26 — archived source and numerical evidence

Files (7): UDCT_v7_14_Step26_Jeans_3D_Link/3d_force_crosscheck.csv, UDCT_v7_14_Step26_Jeans_3D_Link/README.md, UDCT_v7_14_Step26_Jeans_3D_Link/jeans_tail_convergence.csv, UDCT_v7_14_Step26_Jeans_3D_Link/requirements.txt, UDCT_v7_14_Step26_Jeans_3D_Link/results.json, UDCT_v7_14_Step26_Jeans_3D_Link/step24_reference.py, UDCT_v7_14_Step26_Jeans_3D_Link/step26.py

Archived README synopsis: UDCT v7.14 Step 26 — 3D Force / Spherical Jeans Bridge PRE-PREREGISTRATION EXPLORATORY; no new theory claim. Run python step26.py (numpy, scipy). step24_reference.py is copied unchanged from Step 25's reference implementation. The code compares two new 3D finite-box QUMOND solutions (65^3, 97^3, 640 pc cube, zero external field, zero Dirichlet boundary) with the isolated spherical QUMOND analytic acceleration at Plummer scale a=53.3 pc. The finite-box boundary is not certified as isolated. It separately integrates the spherical isotropic Jeans equation for a Plummer tracer with Newtonian stellar-only and isolated spherical QUMOND acceleration. Newtonian analytic radial sigma_r^2(a)=GM/(6 sqrt(2) a) provides a check. Radial dispersion is not projected LOS dispersion. The spherical Jeans integration does not incorporate the 3D computed acceleration directly; the 3D force calculation is a local cross-check only. This is not a 3D Jeans solution and not an observed-star fit. jeans_tail_convergence.csv varies outer radius 1–30 kpc and integration spacing 1 to 0.5 pc. Values near the cutoff are artificial zero-boundary values. 3d_force_crosscheck.csv compares acceleration and reports Hessian eigenvalues, which must NOT be conflated with negative velocity dispersion. results.json preserves constants, output, limitations. Known limitations: spherical isotropy beta=0; no EFE; no selection-function model; no membership likelihood; no Hyd [excerpt]

Archived machine-readable result excerpt: {"status": "EXPLORATORY; no preregistration, no observed-star likelihood", "constants": {"G": 0.00430091, "a_pc": 53.3, "a0_in_(km/s)^2_per_pc": 0.0037028130977896406, "stellar_mass_Msun": 32731.808703363735}, "newtonian_radial_sigma2_analytic_at_a": 0.3112697391849166, "tail_convergence": [{"rmax_pc": 1000, "dr_pc": 1.0, "newton_sig2_at_a": 0.3113547337737417, "qmond_sig2_at_a": 0.38088414542196025, "qmond_sig2_center": 0.5405923271664871, "min_newton_sig2": 0.0863003830568652, "min_qmond_sig2": 0.1947822905281563}, {"rmax_pc": 1000,

"dr_pc": 0.5, "newton_sig2_at_a": 0.3112910466909195, "qmond_sig2_at_a": 0.3808103036488286, "qmond_sig2_center": 0.5408719752360824, "min_newton_sig2": 0.08629717570468, "min_qmond_sig2": 0.19477679272323145}, {"rmax_pc": 3000, "dr_pc": 1.0, "newton_sig2_at_a": 0.3113547903189382, "qmond_sig2_at_a": 0.38088452257866057, "qmond_sig2_center": 0.5405923938235 [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 27 — archived source and numerical evidence

Files (5): README.md, jeans_anisotropy_grid.csv, requirements.txt, results.json, step27.py

Archived README synopsis: UDCT v7.14 — Step 27: Spherical Jeans anisotropy audit Status: pre-preregistration exploratory. Not an observational validation or 3D Jeans solution. Run: python step27.py (requires numpy and scipy). Reproduces jeans_anisotropy_grid.csv and results.json . 48 numerical integrations: Newtonian vs isolated spherical QUMOND, constant beta -0.5, 0, 0.25, 0.5 and Osipkov–Merritt r_a/a 0.5, 1, 2, 5; outer integration limits 3000, 10000, 30000 pc. Grid spacing 0.5 pc. Radial sigma^2 is reported only at 0.5a, a, 2a, 5a (a=53.3 pc). Newtonian beta=0 analytic control at r=a gives GM/(6 sqrt(2) a). Important: Plummer has a core. Constant beta > 0 is not automatically a physically consistent distribution function; central cusp-anisotropy constraints must be tested. Osipkov–Merritt models also need distribution-function positivity checks. Positivity of Jeans sigma^2 alone is necessary but insufficient. No line-of-sight projection, actual individual Hydrus I star likelihood, external field, 3D force integration, or v7.13 B01–B24 replay is included. Previous Step26 is a distinct 3D force cross-check; this step is a spherical Jeans anisotropy extension.

Archived machine-readable result excerpt: {"scope": "Spherical Plummer tracer; Newtonian and isolated spherical QUMOND analytic radial force; exploratory anisotropic Jeans integration", "constants": {"G": 0.00430091, "a_pc": 53.3, "a0_km2_s2_pc": 0.0037028130977896406, "stellar_mass_Msun": 32731.808703363735}, "runs": 48, "newton_isotropic_at_a": 0.31129110330481574, "newton_isotropic_analytic_at_a": 0.3112697391849166, "newton_isotropic_relative_error": 6.863538985535378e-05, "all_sampled_sigma2_positive": true, "cautions": ["Not a direct 3D Jeans solution: spherical radial-force input only", "No external-field effect and no Hydrus I individual-star LOS fit", "Jeans sigma2 positivity does not prove a nonnegative phase-space distribution function", "Constant beta=0.5 can violate central cusp-anisotropy consistency for cored Plummer; do not call physically admissible", "Osipkov-Merritt choices also require DF consistency tests", [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 28 — archived source and numerical evidence

Files (5): README.md, om_distribution_positivity.csv, requirements.txt, results.json, step28.py

Archived README synopsis: UDCT v7.14 — Step 28: Phase-space DF positivity Status: Exploratory, before preregistration. This is a Newtonian Plummer Osipkov–Merritt benchmark, not a QUMOND distribution-function inversion, a 3D Jeans solution, or a fit to individual Hydrus I stars. Reproduce python step28.py with Python, numpy and scipy installed. This creates om_distribution_positivity.csv and results.json . Method For a self-consistent Newtonian Plummer sphere with relative potential Psi(r)=GM/sqrt(r^2+a^2) and OM augmented density rho_Q(r)=(1+r^2/ra^2)rho(r) , define u=Q/Psi0 , q=(a/ra)^2 , Psi0=GM/a . Then rho_Q/rho0=(1-q)u^5+q u^3 . Eddington inversion yields f(Q) = rho0/(sqrt(8) pi^2 Psi0^(3/2)) [(1-q) (128/7) u^(7/2) + 8 q u^(3/2)] . The nonnegative DF criterion for all 0<Q<=Psi0 is exactly q<=16/9 , i.e. ra/a>=3/4 . The script checks 2001 energies and independently reconstructs the augmented density by quadrature at five potential values for each model. Negative DF at ra/a=0.5 and positive DF at ra/a>=0.75 are not new physical discoveries; this is a code and physical-admissibility benchmark. For a cored Plummer tracer, the central cusp-slope anisotropy theorem excludes constant positive central beta (gamma0=0 implies beta0<=0) under its assumptions. Tangential constant beta values are not certified by this test. Interpretation and limitations - The Step27 isolated spherical QUMOND Jeans calculations are not validated as phase-space consistent [excerpt]

Archived machine-readable result excerpt: {"title": "UDCT v7.14 Step28 Newtonian Plummer OM DF admissibility audit", "scope": "Self-consistent isotropic/Osipkov-Merritt Newtonian Plummer only; analytic Eddington inversion plus numerical density reconstruction", "formula": "f(Q)=[rho0/(sqrt(8)*pi^2*Psi0^(3/2))]*[(1-q)*(128/7)*u^(7/2)+8*q*u^(3/2)], u=Q/Psi0, q=(a/ra)^2", "analytic_threshold": "ra/a >= 3/4 for all nonnegative f(Q), 0<Q<=Psi0", "samples": [{"ra_over_a": 0.5, "q": 4.0, "min_fscaled": -22.857142857142854, "fscaled_at_Qmax": -22.857142857142854, "negative_samples": 473, "positive_all_energies": false, "zero_crossing_Q_over_Psi0": 0.7637626158259734, "max_density_reconstruction_abs_error": 3.97326616052851e-12}, {"ra_over_a": 0.7, "q": 2.0408163265306127, "min_fscaled": -2.7055393586005856, "fscaled_at_Qmax": -2.7055393586005856, "negative_samples": 148, "positive_all_energies": false, "zero_crossing_Q_over_Psi0": 0.926 [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 29 — archived source and numerical evidence

Files (7): README.md, necessary_condition.csv, outer_reference_sensitivity.csv, potential_tail.csv, requirements.txt, results.json, step29.py

Archived README synopsis: UDCT v7.14 — Step 29: QUMOND OM DF admissibility preflight Status: exploratory, before preregistration. This package DOES NOT establish QUMOND distribution-function positivity. Reproduce pip install -r requirements.txt then python step29.py . Outputs: necessary_condition.csv , potential_tail.csv , outer_reference_sensitivity.csv , and results.json . Model and equations Spherical Plummer tracer and source, scale a=53.3 pc, luminosity 6546 Lsun, M/L=5 -> stellar mass M=32730 Msun. Isolated QUMOND with simple nu function nu(y)=0.5+sqrt(0.25+1/y) and a0=1.2e-10 m/s^2. The acceleration is g_Q(r)=nu(g_N/a0) g_N where g_N=GM r/(r²+a²)^(3/2) . This spherical algebraic expression is valid in the isolated spherical limit; it is NOT the nonspherical 3D external-field solution. OM augmented tracer density rho_aug(r)=[1+(r/ra)^2]rho(r) . A nonnegative OM DF implies d rho_aug/d Psi >= 0 for the usual finite-escape inversion (and, more generally, monotonicity of the corresponding velocity integral). Since dPsi/dr=-g<0 , this requires d rho_aug/dr<=0 . For Plummer density this is equivalent to ra/a >= sqrt(2/5)=0.6324555 (derivative at the center understood as limit). This is ONLY A NECESSARY CONDITION, not proof of DF positivity. Do not substitute the Newtonian full positivity bound 0.75 for this necessary bound. Critical obstruction to naive Step28 reuse For isolated QUMOND at large r, g_Q(r) ~ sqrt(G M a0)/r ; thus Phi(r)~sqrt(G M a0) log([excerpt]

Archived machine-readable result excerpt: {"title": "UDCT v7.14 Step29 QUMOND DF inversion preflight", "model": {"a_pc": 53.3, "stellar_mass_Msun": 32730.0, "G_pc_kms2_Msun": 0.00430091, "a0_kms2_per_pc": 3.7028130977896407, "nu": "simple", "external_field": 0}, "necessary_OM_condition": "d rho_aug / d Psi >= 0 for nonnegative f(Q), where rho_aug=(1+r\u00b2/ra\u00b2)rho; for Plummer requires ra/a >= sqrt(2/5) ~ 0.63246", "important": "Necessary condition only; passing does NOT establish positive distribution function", "obstruction": "Isolated QUMOND asymptotic potential Phi~sqrt(G*M*a0)*ln(r), diverges as r->infinity; finite-escape relative-potential Eddington inversion from Step28 cannot be reused unchanged", "samples": [{"ra_over_a": 0.5, "min_drho_aug_dPsi": -0.043405365802101, "negative_grid_points": 694, "necessary_condition_pass": false, "sign_change_r_over_a": 0.5}, {"ra_over_a": 0.6, "min_drho_aug_dPsi": -0.006271398641 [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 30 — archived source and numerical evidence

Files (15): README.md, df_positivity_scan.csv, df_q_0p5.csv, df_q_0p6.csv, df_q_0p65.csv, df_q_0p7.csv, df_q_0p75.csv, df_q_1p0.csv, df_q_inf.csv, newtonian_control.csv, newtonian_control.py, requirements.txt, results.json, spot_checks.csv, step30.py

Archived README synopsis: UDCT v7.14 Step 30 — Isolated spherical QUMOND DF inversion (exploratory) This package is an actual numerical run, not preregistration. Execute python step30.py and python newtonian_control.py (NumPy/SciPy). All data in the package are generated, not observational Hydrus I velocities. Assumptions - Plummer mass/tracer density, a=53.3 pc, stellar mass M=32730 solar masses, G=4.30091e-3 pc (km/s)^2/Msun, a0=1.2e-10 m/s². - Isolated spherical QUMOND with simple nu(y)=0.5+sqrt(0.25+1/y), g_Q=nu(g_N/a0)g_N. - Osipkov–Merritt augmented density rho_aug=(1+r²/ra²)rho(r); isotropic ra=infinity. - Positive-energy convention E=v²/2+Phi(r), Phi(0)=0, Phi(infinity)=infinity. For the unbounded potential, rho_aug(Phi) tends to zero as Phi tends to infinity. Method Abel inversion f(E) = [1/(sqrt(8) pi²)] integral_E^infinity rho_aug''(Phi)/sqrt(Phi-E) dPhi. Integrate over radius after substitution ln(r/r_E)=t² to eliminate endpoint square-root singularity. rho_aug''(Phi)=[rho_aug''(r) g(r)-rho_aug'(r) g'(r)]/g(r)^3. Phi(r) integrated on 16001 log-radius points from r/a=1e-7 to 1e8 and interpolated monotonically. Results and interpretation - df_positivity_scan.csv : 11 anisotropy conditions x 43 energies; negative f values appear in all 11 scans, including isotropic case. These are findings for this specific model , not general no-go statements for MOND, QUMOND, or UDCT. - df_q_ .csv : selected full DF curves. - newtonian_control.csv : numerica [excerpt]

Archived machine-readable result excerpt: {"model": {"a_pc": 53.3, "M_Msun": 32730.0, "a0_kms2_per_pc": 3.7028130977896407, "nu": "simple", "external_field": 0}, "inversion": "unbounded-potential Abel inversion, E=v\u00b2/2+Phi, Phi(0)=0; f(E)=1/(sqrt(8) pi\u00b2) integral_E^infinity d\u00b2rho_aug/dPhi\u00b2 /sqrt(Phi-E) dPhi", "samples": [{"ra_over_a": 0.5, "n_energy": 43, "min_df": -0.0019750387878247374, "max_df": 0.002773173822334719, "min_df_over_maxabs": -0.7121943716322707, "negative_energy_points": 10, "first_negative_r_over_a": 0.07815010964380367, "grid_nonnegative": false}, {"ra_over_a": 0.6, "n_energy": 43, "min_df": -0.000985167071764822, "max_df": 0.000705605574052948, "min_df_over_maxabs": -1.0,

"negative_energy_points": 17, "first_negative_r_over_a": 0.014999999999999998, "grid_nonnegative": false}, {"ra_over_a": 0.6324555320336759, "n_energy": 43, "min_df": -0.0008044136936525595, "max_df": 0.000675872601766625 [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 31 — archived source and numerical evidence

Files (7): README.md, density_reconstruction.csv, requirements.txt, results.json, step30_reference.py, step31.py, tail_and_tolerance.csv

Archived README synopsis: UDCT v7.14 Step 31 — QUMOND DF sign and forward reconstruction audit Exploratory, not preregistration. Actual numerical calculations. Source is the locally preserved Step 30 step30.py , included here as step30_reference.py for portable reproduction. Execute python step31.py with NumPy/SciPy. Outputs tail_and_tolerance.csv , density_reconstruction.csv , and results.json . Test - Isolated spherical QUMOND, simple nu, Plummer stellar source and tracer, a=53.3 pc, M=32730 Msun. - Isotropic and Osipkov–Merritt r_a=a. - Compare unbounded-potential DF inversion integrals terminated at r/a=1e4, 1e6, 1e8 and stricter quadrature tolerance. - Forward density reconstruction rho_aug(Phi) = 4 pi sqrt(2) integral_Phi^infty f(E) sqrt(E-Phi) dE using 145 sampled energy points up to r/a=1e6. Findings - At energy corresponding to r/a=0.025, 0.05, 0.1, DF remains negative for both q=inf and q=1. Tail and quadrature sensitivity for those values is less than ~2e-7 relative. - Forward density reconstruction near center (r/a <=0.2) agrees to ~0.08% or better, but error is several percent at r/a=1 to 3 and ~5-11% at r/a=10. Therefore full reconstruction convergence is NOT established. - One integration warning was emitted by scipy for one sampled integral; retain it in the audit trail. - These findings concern a particular imposed tracer profile and potential; do not infer QUMOND/MOND/UDCT general failure, 3D EFE behavior, or real Hydrus I fit. Reprod [excerpt]

Archived machine-readable result excerpt: {"source_step30": "/mnt/data/UDCT_v7_14_Step30_QUMOND_DF_Inversion/step30.py", "audit": "tail 1e4/1e6/1e8 and quadrature tolerance; forward density reconstruction using sampled DF", "tail_and_tolerance": [{"q": "inf", "r_over_a": 0.025, "E": 0.06056610798725432, "f_tail_1e4": -0.004668859500130536, "f_tail_1e6": -0.004668859487051844, "f_tail_1e8": -0.004668858612572882, "f_tight": -0.004668859470876699, "rel_tail_change": 1.8730037358126404e-07, "rel_tolerance_change": 1.8383586446806823e-07}, {"q": "inf", "r_over_a": 0.05, "E": 0.17168961758065898, "f_tail_1e4": -0.0026730157462840818, "f_tail_1e6": -0.0026730151466690705, "f_tail_1e8": -0.002673015209444798, "f_tight": -0.0026730157362927997, "rel_tail_change": 2.3484986955145043e-08, "rel_tolerance_change": 1.9709872811583676e-07}, {"q": "inf", "r_over_a": 0.1, "E": 0.486376410016744, "f_tail_1e4": -0.0009156494354252036, "f_tail_1e6 [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 32 — archived source and numerical evidence

Files (6): README.md, density_convergence.csv, requirements.txt, results.json, step30_reference.py, step32.py

Archived README synopsis: UDCT v7.14 Step 32 — exploratory DF reconstruction convergence This is a pre-preregistration numerical audit , not a claim of observational validation or a general no-go theorem. Model Isolated spherical QUMOND simple interpolation, Plummer stellar source and tracer, scale radius 53.3 pc, stellar mass 5 × 6546 solar masses. Compare isotropic and Osipkov–Merritt anisotropy radius r_a=a. The gravitational potential increases logarithmically at large radius; energy convention E=v²/2+Phi, Phi(0)=0. Method Run python step32.py (requires numpy and scipy). It evaluates the Abel-inverted DF at 145, 289, and 577 logarithmically spaced radii from 0.01a to 1e8a and forward-integrates the sampled DF against sqrt(E-Phi) to reconstruct the augmented tracer density. step30_reference.py preserves original Step30 formulas. Interpretation The density reconstruction errors decrease as energy grid resolution increases. Negative DF samples persist. This is an exploratory self-consistency check, not independent confirmation , because forward and inverse calculations share the same potential and derivatives. No complete 3D external field, Hydrus I star-by-star likelihood, or original v7.13 24-candidate sign test is performed. Known limitations Finite energy range, finite radial extent, numerical quadrature, sparse outer-grid energy samples; positivity cannot be certified globally. The DF sign must be tested with an independent inversion and distribu [excerpt]

Archived machine-readable result excerpt: {"method": "QUMOND OM DF inversion; forward density trapezoid over energy samples, Emax corresponds to r/a=1e8", "rows": [{"q": "isotropic", "N": 145, "r_over_a": 0.05, "reconstructed": 0.9948626326410831, "target": 0.9937772415622736, "relative_error": 0.00109218749777682, "negative_df_grid": 17}, {"q": "isotropic", "N": 145, "r_over_a": 0.1, "reconstructed": 0.9764657788798482, "target": 0.9754310265758153, "relative_error": 0.0010608154506478696, "negative_df_grid": 17}, {"q": "isotropic", "N": 145, "r_over_a": 0.2, "reconstructed": 0.9070879265569328, "target":

0.906601956075185, "relative_error": 0.0005360351127540571, "negative_df_grid": 17}, {"q": "isotropic", "N": 145, "r_over_a": 0.5, "reconstructed": 0.5682548800577408, "target": 0.5724334022399462, "relative_error": -0.00729957784757968, "negative_df_grid": 17}, {"q": "isotropic", "N": 145, "r_over_a": 1.0, "reconstructed": 0. [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 33 — archived source and numerical evidence

Files (6): README.md, central_sign_check.csv, requirements.txt, results.json, step30_reference.py, step33.py

Archived README synopsis: UDCT v7.14 Step 33 — Independent Central Asymptotic Sign Audit Status: Exploratory, pre-preregistration. This is NOT a direct Hydrus I stellar-velocity fit and NOT a general no-go theorem for QUMOND. Scope Isolated spherical simple-QUMOND gravitational field generated by a Plummer stellar mass profile. The tracer is the same Plummer profile, assumed isotropic and in stationary equilibrium. No external field, no 3D anisotropic equilibrium. Independent analytic check For r << a, g_N ~ (G M/a^3) r and simple-QUMOND g ~ sqrt(a0 G M/a^3) sqrt(r). Hence Phi(r)-Phi(0) ~ K r^(3/2), K=(2/3)sqrt(a0 G M/a^3). Plummer rho/rho0 ~ 1 - (5/2)(r/a)^2, giving rho/rho0 ~ 1-B Phi^(4/3), B=5/(2 a² K^(4/3)). The unbounded-potential isotropic Abel inversion is f(E)=1/(sqrt(8) pi²) integral_E^infinity rho''(Phi)/sqrt(Phi-E) dPhi. Since rho'' ~ -(4/9)B Phi^(-2/3), the central contribution is f(E) ~ -4 B Beta(1/2,1/6)/(9 sqrt(8) pi²) E^(-1/6), E->0+. The coefficient is strictly negative, provided the stated model assumptions hold. The outer integral contributes less singular terms. Run python step33.py (requires numpy and scipy). Produces central_sign_check.csv and results.json. step30_reference.py is the archived numerical inversion implementation; step33.py uses its functions only for the numerical side of the comparison. The analytic argument above does not use that numerical inversion. Limitations - The numerical values do not yet agree quantitativ [excerpt]

Archived machine-readable result excerpt: {"model": "isolated spherical simple-QUMOND; Plummer self-consistent stellar tracer; no external field", "analytic_derivation": {"g_small_r": "sqrt(a0 G M/a^3) r^(1/2)", "Phi_small_r": "K r^(3/2)", "rho_small_Phi": "rho0 [1 - B Phi^(4/3)]", "d2rho_dPhi2": "-(4/9) B Phi^(-2/3)", "df_small_E": "-(4 B/(9 sqrt(8) pi^2)) Beta(1/2,1/6) E^(-1/6)", "K": 0.039114429749681, "B": 0.06627802928022407, "leading_coefficient": -0.0076882700941727005}, "rows": [{"r_over_a": 0.0002, "E_kms2": 4.307594079717392e-05, "numerical_f": -0.03339335430797165, "asymptotic_f": -0.04106351956177551, "ratio_numeric_to_asymptotic": 0.8132121811364721, "sign_agrees": true}, {"r_over_a": 0.0004, "E_kms2": 0.00012186900121342507, "numerical_f": -0.026868234315503583, "asymptotic_f": -0.034528662564260876, "ratio_numeric_to_asymptotic": 0.7781429201174368, "sign_agrees": true}, {"r_over_a": 0.0008, "E_kms2": 0.0003448223 [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 34 — archived source and numerical evidence

Files (6): README.md, central_om_comparison.csv, requirements.txt, results.json, step30_reference.py, step34.py

Archived README synopsis: UDCT v7.14 exploratory Step 34 — central OM DF sign audit This is pre-preregistration exploratory work , not a UDCT validation or a direct fit to Hydrus I stars. Model Isolated spherical Plummer stellar mass and tracer, M=32730 Msun, a=53.3 pc; simple-QUMOND acceleration g = gN/2 + sqrt(gN²/4 + a0 gN) , a0=1.2e-10 m/s². OM augmented density rho_aug=(1+r²/ra²) rho_Plummer . Potential defined with Phi(0)=0 and diverges logarithmically at infinity. Numerical inversion core is the archived Step30 code, copied unchanged. Local derivation As r->0, g ~ sqrt(a0 GM/a³) r^(1/2), so Phi ~ K r^(3/2), K=(2/3)sqrt(a0 GM/a³). Normalize rho0=1. With q=ra/a, rho_aug=1+(q^-2-5/2)(r/a)²+O(r^4). Thus rho_aug(Phi)=1+D Phi^(4/3)+... where D=(q^-2-5/2)/(a² K^(4/3)). The unbounded-potential Abel inversion yields f(Q)~ [4 Beta(1/2,1/6)/(9 sqrt(8) pi²)] D Q^(-1/6). The sign of the leading central term flips at q=sqrt(2/5). At equality the leading term vanishes and higher-order terms are necessary. Run pip install -r requirements.txt python step34.py Outputs: central_om_comparison.csv , results.json . Interpretation and limits 54 direct numerical comparisons across 9 anisotropy choices and 6 inner radii. Numerical comparison uses the same Step30 inversion implementation: analytic local derivation is independent, but there is not a second independent numerical inversion algorithm. Positive central leading coefficient does NOT establish global DF positivi [excerpt]

Archived machine-readable result excerpt: {"model": "isolated spherical simple-QUMOND with Plummer stellar tracer and source", "units": "pc, km/s, Msun; normalized tracer density", "central_exponent": "f(E) proportional to E^(-1/6)", "critical_q": 0.6324555320336759, "leading_coefficient_prefactor": 0.0030753080376690806, "number_of_numerical_checks": 54, "checks": [{"q": 0.5, "r_over_a": 1e-05, "E_kms2": 4.808983807262545e-07, "numerical_f": 0.046473564660154246, "asymptotic_f": 0.05211599054099163, "ratio": 0.8917333082943256, "sign_agreement": true}, {"q": 0.5, "r_over_a": 3e-05, "E_kms2":

2.501092184387727e-06, "numerical_f": 0.0339632196650216, "asymptotic_f": 0.03959359464640946, "ratio": 0.857795811881444, "sign_agreement": true}, {"q": 0.5, "r_over_a": 0.0001, "E_kms2": 1.5226597479463714e-05, "numerical_f": 0.02367828880774576, "asymptotic_f": 0.0293007950378758, "ratio": 0.8081107962134788, "sign_agreement": true}, {"q [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 35 — archived source and numerical evidence

Files (5): README.md, independent_gauss_df.csv, requirements.txt, results.json, step35.py

Archived README synopsis: UDCT v7.14 Step 35 — Independent quadrature audit of OM DF signs Status: exploratory, pre-preregistration. Not a UDCT-specific prediction, 3D external-field test, or direct Hydrus I fit. Scope Independent numerical integration method for the unbounded-potential Abel inversion of a spherical, isolated, simple-QUMOND Plummer tracer and source. Stellar mass M=32730 Msun, Plummer a=53.3 pc, a0=1.2e-10 m/s², zero external field. The OM augmented tracer density is rho_aug=(1+r²/ra²)rho_Plummer. The independent step35.py uses fixed-order Gauss-Legendre quadrature in t=sqrt(log(r/r_E)), and direct adaptive quadrature for the potential difference Phi(r)-Phi(r_E). It does not import or call the Step30 df() or its interpolated potential. Both approaches share the analytic QUMOND acceleration and tracer derivatives; this is an independent integration algorithm, not independent physics. Results - 7 OM/anisotropy cases x 11 turning radii = 77 inversion evaluations, each with 96- and 192-point Gauss quadrature. - Maximum relative discrepancy (96 vs 192 nodes): about 0.0137%. - All seven cases contain sampled negative DF values. q=0.5 and 0.6 are positive in the central sample yet negative at intermediate turning radii. - At q=sqrt(2/5), the local x² term in rho_aug cancels, but the full inversion is numerically negative at the sampled center, ~-0.0006346. The local x^4 term is -15/8 x^4. This local coefficient does NOT determine the sign of [excerpt]

Archived machine-readable result excerpt: {"method": "independent Gauss-Legendre quadrature with direct quadrature of potential difference; does not call step30 inversion or its PCHIP potential", "model": {"a_pc": 53.3, "M_Msun": 32730.0, "a0_kms2_per_pc": 3.7028130977896407, "external_field": 0}, "q_critical": 0.6324555320336759, "rho_aug_x4_at_critical": -1.875, "n_evaluations": 77, "max_relative_96_192": 0.00013688581895453822, "critical_rows": [{"q": 0.6324555320336759, "r_over_a": 1e-05, "df_gauss96": -0.0006345193475164711, "df_gauss192": -0.0006346062161080767, "relative_delta": 0.00013688581895453822, "sign": "negative"}, {"q": 0.6324555320336759, "r_over_a": 0.0001, "df_gauss96": -0.0006346265930858828, "df_gauss192": -0.0006346159355512559, "relative_delta": 1.6793676348026132e-05, "sign": "negative"}, {"q": 0.6324555320336759, "r_over_a": 0.001, "df_gauss96": -0.000634891956121354, "df_gauss192": -0.000634892402140615 [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 36 — archived source and numerical evidence

Files (5): README.md, requirements.txt, results.json, step36.py, tracer_scale_scan.csv

Archived README synopsis: UDCT v7.14 Step 36 — tracer-scale sensitivity (exploratory) Run: python step36.py (requires numpy and scipy). Output: tracer_scale_scan.csv and results.json. Fixed baryonic Plummer mass M=32730 Msun and scale a=53.3 pc; isotropic Plummer tracer scale b/a = 0.5,0.75,1,1.5,2. The tracer is normalized to unit central density, so reported DF amplitudes are dimensionless-normalized and only sign comparisons are intended. Center-referenced potential, isolated spherical simple-QUMOND g = (gN+sqrt(gN^2+4 a0 gN))/2. Integration uses r=r0 exp(t²) and direct potential quadrature. All 25 tests are exploratory, not preregistered. The sampled central energies yield negative DF for all tracer scales, but there is no all-energy proof, no external field, no 3D equilibrium, no individual-star Hydrus I fit, and no UDCT-specific physics. A negative isotropic DF for this specified tracer/potential pair does not falsify QUMOND generally. Step 35 and earlier data are not embedded; this script is self-contained.

Archived machine-readable result excerpt: {"cases": 25, "negative": 19, "positive": 6, "model": "Isolated spherical simple QUMOND, fixed Plummer gravitating mass profile and variable Plummer isotropic tracer scale", "a_pc": 53.3, "M_msun": 32730, "limitations": ["25 sampled energies only", "No external field", "Not a full 3D Jeans fit", "Not a test of UDCT-specific terms", "No Hydrus I individual-star likelihood"]}

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 37 — archived source and numerical evidence

Files (6): README.md, cusp_profile_scan.csv, requirements.txt, results.json, run.log, step37.py

Archived README synopsis: UDCT v7.14 Step 37 — Central tracer-shape sensitivity (exploratory) Run python step37.py (Python 3, NumPy, SciPy). Outputs cusp_profile_scan.csv and results.json . Setup Fixed baryonic Plummer source: M=32730 Msun, a=53.3 pc, isolated spherical simple-QUMOND acceleration. Normalized stellar tracer density: rho(r)/rho(0)=[1+(r/a)^p]^(-5/p). The tracer profile is finite at r=0 for all p. Here 'central shape' is more accurate than 'cusp': p<2 has a nonanalytic center, but density does not diverge. The distribution function is computed via a center-referenced, unbounded-potential isotropic Abel inversion. p=2 recovers Plummer tracer. This is a controlled model-family sensitivity, not a realistic profile fit. Findings 40 evaluations (8 p values x 5 energy turning radii). p=1,1.25,1.5 are positive at all sampled points; p>=1.75 has at least one negative sampled value. The sign at small energies depends strongly on the tracer center-shape assumption. This does not imply all-energy positivity for p<=1.5. Limits No external field; no anisotropic 3D equilibrium; no full Hydrus I individual-star fit; no claim about UDCT-specific gravity or QUMOND generally. Values normalized to unit central tracer density; compare signs, not amplitudes across p.

Archived machine-readable result excerpt: {"step": 37, "n_cases": 40, "negative": 20, "positive": 20, "p_values": [1.0, 1.25, 1.5, 1.75, 2.0, 2.5, 3.0, 4.0], "r_over_a": [0.0001, 0.001, 0.01, 0.1, 0.5], "model": "fixed isolated spherical simple-QUMOND gravity from Plummer baryonic mass; tracer rho=(1+(r/a)^p)^(-5/p)", "note": "The density family has a finite central value for all p, with different approach-to-center exponents; not a divergent density cusp.", "cautions": ["Sampled energies only, not global DF positivity", "No external field or 3D DF", "No direct Hydrus I observational fit", "Numerical quadrature error is not a physical systematic error bound"]}

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 38 — archived source and numerical evidence

Files (5): UDCT_v7_14_Step38_Full_Energy_Audit/README.md, UDCT_v7_14_Step38_Full_Energy_Audit/full_energy_scan.csv, UDCT_v7_14_Step38_Full_Energy_Audit/requirements.txt, UDCT_v7_14_Step38_Full_Energy_Audit/results.json, UDCT_v7_14_Step38_Full_Energy_Audit/step38.py

Archived README synopsis: UDCT v7.14 — Step 38: expanded energy and central-shape audit Status: exploratory, not preregistration, not an observational fit. Model - Spherical isolated simple-QUMOND acceleration from Plummer baryonic mass M=32730 solar masses, a=53.3 pc, a0=1.2e-10 m/s². - Isotropic tracer density normalized to unity at center: rho=(1+(r/a)^p)^(-5/p). - Center-referenced energy inversion in an unbounded logarithmic asymptotic potential; the tabulated r is the energy-turning radius, not an observed star position. - Log-radius potential integration and PCHIP interpolation; Gauss–Legendre integration with 256 and 512 nodes for a numerical comparison. Reproduce python step38.py requires NumPy and SciPy. Regenerates full_energy_scan.csv and results.json . Result 13 p values times 91 log-spaced turning radii in [1e-6 a, 1e3 a]. All 91 sampled values nonnegative for p<=1.5; negative values appear at p=1.55 and higher. The observed onset is bounded by the tested grid (1.50, 1.55], not a proven physical critical exponent. Numerical limitations The difference between 256- and 512-node estimates is a quadrature convergence check only. Both use the same interpolated potential and derivative formulas, so systematic bias is shared. For p=1 the largest absolute N-difference is ~1.96e-6; this deserves targeted refinement. Potential interpolation starts at r/a=1e-10, not r=0; finite outer radius r/a=1e9 for inversion; far-tail and center asymptotics are [excerpt]

Archived machine-readable result excerpt: {"model": "fixed Plummer gravity, isotropic generalized cored tracer, isolated simple QUMOND", "sampled_r_over_a": [1e-06, 1000.0], "n_per_p": 91, "n": 1183, "by_p": {"1.0": {"negative_count": 0, "first_negative_r_over_a": null, "last_negative_r_over_a": null, "max_abs_N_difference": 1.9563522073440254e-06}, "1.25": {"negative_count": 0, "first_negative_r_over_a": null, "last_negative_r_over_a": null, "max_abs_N_difference": 3.0942146622692235e-08}, "1.4": {"negative_count": 0, "first_negative_r_over_a": null, "last_negative_r_over_a": null, "max_abs_N_difference": 1.5587602320010774e-09}, "1.5": {"negative_count": 0, "first_negative_r_over_a": null, "last_negative_r_over_a": null, "max_abs_N_difference": 1.1868978022633314e-13}, "1.55": {"negative_count": 39, "first_negative_r_over_a": 1e-06, "last_negative_r_over_a": 0.006309573444801936, "max_abs_N_difference": 9.805889433778248e-11}, [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 39 — archived source and numerical evidence

Files (5): UDCT_v7_14_Step39_Critical_Exponent/README.md, UDCT_v7_14_Step39_Critical_Exponent/critical_scan.csv, UDCT_v7_14_Step39_Critical_Exponent/requirements.txt, UDCT_v7_14_Step39_Critical_Exponent/results.json, UDCT_v7_14_Step39_Critical_Exponent/step39.py

Archived README synopsis: UDCT v7.14 — Step 39: Critical exponent audit (exploratory) Scope Follow-up to Step 38. Reuses its explicit spherical simple-QUMOND acceleration prescription and generalized cored tracer density, but runs a new focused 81-point numerical grid. This is not a fit to individual Hydrus I velocities, not a 3D external-field calculation, and not a preregistered result. Model - Fixed Plummer gravitating mass: M=32730 solar masses, a=53.3 pc; G=4.30091e-3 pc (km/s)^2 / solar mass. - Isolated spherical simple-QUMOND acceleration: g=(g_N/2)+sqrt(g_N^2/4+a0 g_N), a0=1.2e-10 m/s^2. - Isotropic tracer density rho=(1+(r/a)^p)^(-5/p) (normalization rho0=1). - Potential increases outward, Psi(r)=Phi(r)-Phi(0). Energy E=Psi(r0). The inversion is the unbounded-potential Abel inversion f(E)=1/(sqrt(8) pi^2) integral_E^infinity rho''(Psi)/sqrt(Psi-E) dPsi, with boundary assumptions implicit in the formula. The integral is transformed to log radius and regularized near the lower endpoint. Asymptotic prediction At small r: g_N~r, g~r^(1/2), Psi~r^(3/2). The tracer behaves as rho~rho0-A Psi^(2p/3). For p != 1.5, the leading singular contribution to f(E) is proportional to (1-2p/3) E^(2p/3-3/2) with positive prefactor in the applicable asymptotic regime (p near 1.5). Thus p=1.5 is a central asymptotic sign boundary. At p=1.5 the leading coefficient cancels and subleading/nonlocal terms must be retained. Numerical verification python step39.py writes [excerpt]

Archived machine-readable result excerpt: {"model": "isolated spherical simple-QUMOND fixed Plummer gravity; isotropic generalized cored tracer", "critical_asymptotic_p": 1.5, "sample_count": 81, "radius_samples": [1e-06, 1e-05, 0.0001, 0.001, 0.01, 0.1, 1, 10, 100], "p_samples": [1.45, 1.49, 1.499, 1.5, 1.501, 1.505, 1.51, 1.525, 1.55], "counts": {"1.45": 0, "1.49": 0, "1.499": 0, "1.5": 0, "1.501": 2, "1.505": 3, "1.51": 3, "1.525": 4, "1.55": 4}, "caveats": ["finite radius grid does not establish global positivity", "the p=1.5 leading singular coefficient vanishes and subleading terms matter", "quadrature convergence does not validate physical assumptions"]}

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 40 — archived source and numerical evidence

Files (5): UDCT_v7_14_Step40_Asymptotic_Audit/README.md, UDCT_v7_14_Step40_Asymptotic_Audit/audit.csv, UDCT_v7_14_Step40_Asymptotic_Audit/requirements.txt, UDCT_v7_14_Step40_Asymptotic_Audit/step40.py, UDCT_v7_14_Step40_Asymptotic_Audit/summary.json

Archived README synopsis: UDCT v7.14 — Step 40: Critical-exponent asymptotic audit Status: exploratory, NOT preregistered, not a Hydrus I star-by-star fit, not 3D EFE. Reproduction python -m pip install -r requirements.txt python step40.py This produces audit.csv (30 samples) and summary.json . The numerical calculation reuses the physical model and inversion equation of Step 39 but recomputes the potential on a denser, wider grid and evaluates 192- and 384-node Gauss–Legendre integrals. Thus it is an independently rerun numerical calculation, not an independent proof of the inversion theorem . The small differences between node counts assess only quadrature convergence. Model G=4.30091e-3 pc (km/s)^2/Msun; M=32730 Msun; a=53.3 pc; a0=1.2e-10 m/s². Plummer gravitating mass with isolated spherical simple-QUMOND acceleration gN(r)=GM r/(r²+a²)^(3/2) g(r)=gN/2 + sqrt(gN²/4+a0 gN) Tracer density rho_p(r)=[1+(r/a)^p]^(-5/p), with rho(0)=1. Potential Psi=Phi(r)-Phi(0), E=Psi(r0), and the unbounded-potential isotropic Abel inversion f(E) = [1/(sqrt(8) pi²)] integral_E^infinity rho''(Psi)/sqrt(Psi-E) dPsi. The inversion assumes adequate decay at infinity and a stationary isotropic DF f(v²/2+Psi); it does not establish physical stability or uniqueness beyond these assumptions. New asymptotic calculation Let k=GM/a³ and h=sqrt(a0 k). As r→0: g(r)=h r^(1/2)+(k/2)r+O(r^(3/2)) Psi(r)=A r^(3/2)+(k/4)r²+O(r^(5/2)), A=2h/3. For p near 3/2, set alpha=2p/3. Then rho(Psi [excerpt]

Archived machine-readable result excerpt: {"k": 0.0009296612580853805, "h": 0.0586716446245215, "A": 0.039114429749681, "B": 0.21900343602734065, "Dcoef": 0.003833537061489486, "critical_coefficient": 0.00044469138061029365, "results": [{"p": 1.49, "r_over_a": 0.0001, "E": 1.5227065691232654e-05, "f192": 0.055313492447964636, "f384": 0.05531349244775056, "relative_difference": 3.8702809251389406e-12, "leading_asymptotic": 0.04574810906563048, "critical_subleading": null}, {"p": 1.49, "r_over_a": 1e-05, "E": 4.813792867894178e-07, "f192": 0.27525408859922806, "f384": 0.27525408860028966, "relative_difference": 3.856782878484943e-12, "leading_asymptotic": 0.2632924572664668, "critical_subleading": null}, {"p": 1.49, "r_over_a": 1e-06, "E": 1.5221121830168048e-08, "f192": 1.531483468575598, "f384": 1.5314834685813654, "relative_difference": 3.76588235305287e-12, "leading_asymptotic": 1.5151619584003162, "critical_subleading": null} [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 41 — archived source and numerical evidence

Files (6): README.md, comparison.csv, independent_adaptive.csv, requirements.txt, step41.py, summary.json

Archived README synopsis: UDCT v7.14 — Step 41: Independent quadrature audit Status: pre-preregistration exploratory numerical audit, 2026-10-25. What was independently changed Step 40 used a tabulated/interpolated potential and Gauss–Legendre quadrature. This step computes potential differences by adaptive scipy.integrate.quad on every integrand evaluation, and independently evaluates the outer inversion integral using adaptive QUAD. No potential interpolation or Legendre nodes are used. The same underlying isotropic Abel inversion equation and the same spherical Plummer-QUMOND physical assumptions are retained; this is NOT an independent derivation of that equation. Model G=4.30091e-3 pc (km/s)^2/Msun; M=32730 Msun; a=53.3 pc; a0=1.2e-10 m/s^2. gN=G M r/(r r+a a)^1.5; g=gN/2+sqrt(gN gN/4+a0 gN). Tracer rho=(1+(r/a)^p)^(-5/p), isotropic DF; potential Phi(r)-Phi(0) unbounded as r tends to infinity. f(E) = 1/(sqrt(8) pi^2) integral_E^infty rho''(Psi)/sqrt(Psi-E) dPsi. The integral is numerically truncated at r=1e6 a, with the tail not rigorously bounded in this step. Run pip install -r requirements.txt python step41.py Outputs: independent_adaptive.csv, summary.json. comparison.csv compares against Step 40 (precomputed, included in ZIP). 21 calculations: 7 exponents by 3 radii. The calculations agree closely with Step 40, supporting numerical implementation robustness rather than independent physical validation. Cautions A local center asymptotic sign b [excerpt]

Archived machine-readable result excerpt: {"method": "adaptive scipy QUAD with potential integrated on demand; no PCHIP; independent quadrature, same analytical inversion identity", "rows": [{"p": 1.49, "r_over_a": 0.0001, "f_adaptive": 0.05531349280616654, "quad_error_est": 2.6648634534324764e-12, "sign": "positive"}, {"p": 1.49, "r_over_a": 1e-06, "f_adaptive": 1.531483478664897, "quad_error_est": 5.325634573572877e-09, "sign": "positive"}, {"p": 1.49, "r_over_a": 1e-08, "f_adaptive": 50.211413622180125, "quad_error_est": 6.187731496136262e-08, "sign": "positive"}, {"p": 1.499, "r_over_a": 0.0001, "f_adaptive": 0.013553625041341335, "quad_error_est": 2.1112265267460742e-11, "sign": "positive"}, {"p": 1.499, "r_over_a": 1e-06, "f_adaptive": 0.15038599373729092, "quad_error_est": 4.780794570839916e-09, "sign": "positive"}, {"p": 1.499, "r_over_a": 1e-08, "f_adaptive": 4.321045926663445, "quad_error_est": 4.148130793805234e-08, " [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 42 — archived source and numerical evidence

Files (5): README.md, density_reconstruction.csv, requirements.txt, step42.py, summary.json

Archived README synopsis: UDCT v7.14 Step 42 — Forward density reconstruction audit Status: exploratory, pre-preregistration. 2026-10-25. Aim Test whether the signed distribution function recovered using the right-sided Abel inversion reconstructs the input tracer density. This is an internal consistency check, not a proof of physical DF positivity, observational fit, or a general failure of QUMOND. Model Spherical isolated QUMOND with Plummer gravitating mass M=32730 Msun, a=53.3 pc, acceleration scale a0=1.2e-10 m/s^2, simple nu(x)=0.5+sqrt(0.25+1/x). Tracer density rho(r)/rho0=[1+(r/a)^p]^(-5/p). Spherical potential is defined by dPhi/dr=g(r)>0; its zero-point cancels from integrals. In an isolated deep-MOND exterior, Phi diverges logarithmically, so the relevant inversion is the right-sided Abel transform over Phi to +infinity. rho(Phi)=4 pi sqrt(2) integral_Phi^infinity f(E) sqrt(E-Phi) dE f(E)=1/(sqrt(8) pi^2) integral_E^infinity rho''(Phi)/sqrt(Phi-E) dPhi Procedure The Python script evaluates g, dg/dr, rho', rho'', constructs Phi(r) by cumulative quadrature on a logarithmic radial grid, obtains rho''(Phi), performs inverse Abel quadrature with endpoint substitution, then forward-integrates the resulting signed f to recover rho. Compare 900 and 1400 radial grid points; Gauss-Legendre order 112. Outer cutoff rmax/a=1e6. Reconstruction at r/a=1e-4,1e-3,1e-2,0.1,0.5,1,3,10. Four density exponents p=1.49,1.5,1.501,2.0. Results Maximum absolute relat [excerpt]

Archived machine-readable result excerpt: {"cases": [{"p": 1.49, "negative_samples": 0, "positive_samples": 899, "f_min": 0.0, "f_max": 0.17393836354856199, "rmax_over_a": 1000000.0, "ngrid": 900, "nq": 112}, {"p": 1.49, "negative_samples": 0, "positive_samples": 1399, "f_min": 0.0, "f_max": 0.17390295570207878, "rmax_over_a": 1000000.0, "ngrid": 1400, "nq": 112}, {"p": 1.5, "negative_samples": 0, "positive_samples": 899, "f_min": 0.0, "f_max": 0.010938369893892193, "rmax_over_a": 1000000.0, "ngrid": 900, "nq": 112}, {"p": 1.5, "negative_samples": 0, "positive_samples": 1399, "f_min": 0.0, "f_max": 0.01093727924929509, "rmax_over_a": 1000000.0, "ngrid": 1400, "nq": 112}, {"p": 1.501, "negative_samples": 340, "positive_samples": 559, "f_min": -0.003707810430893841, "f_max": 0.007327713184931298, "rmax_over_a": 1000000.0, "ngrid": 900, "nq": 112}, {"p": 1.501, "negative_samples": 529, "positive_samples": 870, "f_min": -0.003705839 [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 43 — archived source and numerical evidence

Files (6): README.md, cutoff_scan.csv, requirements.txt, step42.py, step43.py, summary.json

Archived README synopsis: UDCT v7.14 — Step 43: Outer cutoff sensitivity audit Status: pre-preregistration exploratory computation. 2026-10-25. Scope Reuses the Step 42 inversion implementation unchanged and varies the maximum integration radius. Four tracer density exponents p = 1.49, 1.5, 1.501, 2.0; four outer cutoff radii R/a = 1e4, 1e5, 1e6, 1e7; 1000 log-spaced radial grid points and 128 Gauss-Legendre nodes. Eight reconstruction radii per case, 128 reconstruction comparisons in total. Potential and density assumptions: fixed spherical Plummer baryonic gravitational source with a=53.3 pc, M=32730 Msun, G=4.30091e-3 pc (km/s)^2/Msun; spherical QUMOND algebraic acceleration g = gN/2 + sqrt(gN^2/4 + a0 gN), a0=1.2e-10 m/s^2; isotropic tracer rho=(1+(r/a)^p)^(-5/p). Run From the extracted archive directory, run: python -m pip install -r requirements.txt python step43.py step43.py imports step42.py , included here as a frozen dependency. Outputs cutoff_scan.csv and summary.json . Results At R/a=1e4, 1e5, 1e6, 1e7 respectively, negative grid sample counts: - p=1.49: 0,0,0,0 - p=1.5: 0,0,0,0 - p=1.501: 434,406,382,360 - p=2.0: 678,636,598,565 Max absolute relative density reconstruction errors across the 8 radii for each cutoff are approximately 0.0336%, 0.0272%, 0.0434%, 0.0497%. Results depend slightly on p. Important limitations 1. This is not a rigorous bound on the omitted infinite-radius integral. A bound requires asymptotic tail analysis or a con [excerpt]

Archived machine-readable result excerpt: {"cases": [{"p": 1.49, "cutoff_over_a": 10000.0, "max_abs_relative_density_error": 0.00033635015477789294, "negative_grid_samples": 0, "minimum_df": 0.0}, {"p": 1.49, "cutoff_over_a": 100000.0, "max_abs_relative_density_error": 0.0002723203119641741, "negative_grid_samples": 0, "minimum_df": 0.0}, {"p": 1.49, "cutoff_over_a": 1000000.0, "max_abs_relative_density_error": 0.00043391988965539114, "negative_grid_samples": 0, "minimum_df": 0.0}, {"p": 1.49, "cutoff_over_a": 10000000.0, "max_abs_relative_density_error": 0.0004971356511194214, "negative_grid_samples": 0, "minimum_df": 0.0}, {"p": 1.5, "cutoff_over_a": 10000.0, "max_abs_relative_density_error": 0.000336361956061555, "negative_grid_samples": 0, "minimum_df": 0.0}, {"p": 1.5, "cutoff_over_a": 100000.0, "max_abs_relative_density_error": 0.0002723346855569703, "negative_grid_samples": 0, "minimum_df": 0.0}, {"p": 1.5, "cutoff_over_a [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 44 — archived source and numerical evidence

Files (5): README.md, fixed_energy_tail.csv, requirements.txt, step44.py, summary.json

Archived README synopsis: UDCT v7.14 — Step 44: Fixed-energy outer-tail audit Status: exploratory, not preregistration , 2026-10-25. Scientific question Does the negative isotropic distribution function in a specified isolated spherical Plummer–QUMOND model depend on truncating the outer radial integration? Model and method Gravitating mass: spherical Plummer with a=53.3 pc, M=32730 solar masses. G=4.30091e-3 pc (km/s)^2 / Msun; a0=1.2e-10 m/s^2. Spherical acceleration g=gN/2+sqrt(gN^2/4+a0 gN). Isotropic tracer density rho=(1+(r/a)^p)^(-5/p). p in [1.49,1.5,1.501,2.0]. The isotropic inversion is f(E)=1/(sqrt(8) pi^2) integral_E^infty [d²rho/dPhi²] / sqrt(Phi-E) dPhi, with the boundary derivative at infinity vanishing for the chosen tracer. E=Phi(r_E). We compute Phi(r)-Phi(r_E) by solving dPhi/d(log r)=r g(r) using an adaptive ODE solver. The change of variables log(r/r_E)=t² removes the lower-end Abel square-root singularity. The radial integral is then evaluated using scipy.integrate.quad. No PCHIP interpolation of the potential or Gauss-Legendre quadrature from Step 43 is used. Cutoffs R/a = 1e4,1e5,1e6,1e7,1e9,1e12; reference R/a=1e14 (finite). For each of 4 p and 4 fixed physical energies (r_E/a=1e-4,0.01,0.1,1), 6 cutoff comparisons: 96 rows. Findings At r_E/a=1e-4, reference f values: p=1.49: +0.05531349112; p=1.5: +0.009280164943; p=1.501: +0.005076663140; p=2: -0.04115614022. At p=1.501, f(r_E/a=1e-5) ≈ -0.012001 and f(r_E/a=1e-6) ≈ -0.116173 [excerpt]

Archived machine-readable result excerpt: {"cases": 96, "reference_outer_radius_over_a": 100000000000000.0, "reference_is_finite_not_proven_infinite": true, "max_abs_delta_by_cutoff": {"10000.0": 8.303911464696556e-11, "100000.0": 6.683233827464719e-11, "1000000.0": 6.785093754207994e-11, "10000000.0": 8.324192030118027e-11, "1000000000.0": 1.1650182207834092e-10, "1000000000000.0": 1.2485647932214405e-10}, "min_abs_reference_df": 0.00046453404229980684, "warnings": ["Numerical tail estimate, not a rigorous mathematical bound", "Same spherical QUMOND field and isotropic Abel inversion assumptions", "Fixed physical energy (rE/a), unlike earlier changing-grid sample counts", "Reference 1e14 a is finite; no claim of exact infinity"]}

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 45 — archived source and numerical evidence

Files (6): README.md, om_scan.csv, requirements.txt, step45.py, summary.csv, summary.json

Archived README synopsis: UDCT v7.14 — Step 45: OM Anisotropy Audit Status: pre-preregistration exploration, 2026-10-25. Objective Test whether the central negative distribution function in the isolated spherical Plummer-QUMOND field persists when the generalized cored tracer is assigned Osipkov–Merritt (OM) anisotropy. Model - Gravitating Plummer source: mass M=32730 Msun, scale a=53.3 pc. - Acceleration in spherical QUMOND: g = g_N/2 + sqrt(g_N^2/4 + a0 g_N), with a0=1.2e-10 m/s^2. - Tracer density normalized to unity at center: rho=(1+(r/a)^p)^(-5/p). - OM augmented tracer: rho_Q=(1+r^2/ra^2)rho. - OM inversion of rho_Q with respect to the potential Phi(r) with Phi'(r)=g(r); the zero of Phi is arbitrary and the potential diverges logarithmically at infinity. - Isotropic case ra=infinity. - Formula: f(Q)=(1/(sqrt(8) pi^2)) integral_{Phi(Q)}^infty [d²rho_Q/dPhi²]/sqrt(Phi-Phi(Q)) dPhi, where Q denotes the appropriate OM energy coordinate under this convention. - Numerically integrate in logarithmic radius with a square-root change of variables, solving potential differences with solve_ivp and integrating with scipy.quad. Reproduce Python 3 with numpy and scipy: pip install -r requirements.txt python step45.py Outputs om_scan.csv , summary.csv , and summary.json . Results 4 p values × 5 ra/a values × 6 radius-to-energy mappings = 120 computed points; 16 negative. See summary.csv for per-model counts and om_scan.csv for each value. Notably, at p=1.501, [excerpt]

Archived machine-readable result excerpt: {"n_cases": 120, "n_negative": 16, "model": "spherical isolated QUMOND with generalized cored tracer and OM augmented density", "outer_cutoff_a": 1000000000.0, "notes": ["OM f(Q) is not the same as isotropic f(E)", "finite grid: no claim of positivity over all energies", "no actual Hydrus I individual star velocity fit", "common potential and inversion assumptions"]}

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 46 — archived source and numerical evidence

Files (6): README.md, om_2d_scan.csv, requirements.txt, sign_map_summary.csv, step46.py, summary.json

Archived README synopsis: UDCT v7.14 — Step 46: OM anisotropy sign map Status: exploratory, before preregistration . No Hydrus I individual-star likelihood fitting. Purpose Map signs of an Osipkov–Merritt (OM) distribution function in an isolated spherical QUMOND potential sourced by a Plummer mass distribution. Tracer density: rho(r) = [1+(r/a)^p]^(-5/p). OM augmented density: (1+r^2/r_a^2) rho(r). Computation python step46.py regenerates om_2d_scan.csv , sign_map_summary.csv , and summary.json . - 11 p values: 1.45, 1.48, 1.49, 1.495, 1.5, 1.501, 1.505, 1.51, 1.55, 1.6, 2.0 - 10 anisotropy radii r_a/a: 0.4, 0.5, 0.6, 0.65, 0.75, 1.0, 1.5, 2.0, 4.0, infinity (isotropic) - 33 log-spaced energy-associated turning radii from 1e-7 a to 1e2 a - 3630 f(Q) values, of which 800 are negative - 96- and 192-point Gauss–Legendre quadrature. No sign disagreements; largest absolute difference 2.87275e-5 in the chosen normalized units. Method The force g(r) and derivative dg/dr are analytic; potential differences are computed by a high-accuracy ODE integration over log radius. The OM augmented density derivatives are analytic. The Eddington/OM Abel inverse integral is computed over log radius with a square-root-endpoint-removing substitution. Outer numerical radius is 1e9 a. r_energy_over_a labels the radial point where Q=Phi(r), not an observed star radius. Interpretation and limitations - p=1.5 is not a universal positivity boundary for anisotropic models. For r_a [excerpt]

Archived machine-readable result excerpt: {"cases": 3630, "negative_cases": 800, "max_abs_quadrature_difference": 2.8727503739389704e-05, "n_p": 11, "n_ra": 10, "n_energy": 33, "caveat": "Finite sampled energies; zero negatives does not certify globally nonnegative DF; same analytic gravity and OM inversion used at both quadrature orders."}

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 47 — archived source and numerical evidence

Files (7): README.md, exact_om_zero_contour.csv, refined_energy_zero_crossings.csv, requirements.txt, step46_om_2d_scan_input.csv, step47.py, summary.json

Archived README synopsis: UDCT v7.14 — Step 47: OM zero-contour audit Status: exploratory, not preregistered , 2026-10-25. Reproduction 1. Install pip install -r requirements.txt . 2. Place the original Step 46 om_2d_scan.csv at ../UDCT_v7_14_Step46_OM_Sign_Map/om_2d_scan.csv and run python step47.py . 3. The ZIP also includes a frozen copy of Step 46 input as step46_om_2d_scan_input.csv . To run from the ZIP alone, the script automatically falls back to this local copy. Mathematics For the fixed spherical gravitational potential and the Osipkov–Merritt augmented tracer density, rho_OM(r) = rho(r) [1 + (r/r_a)^2] , the DF inversion is linear. With r_a/a dimensionless, f(Q;r_a) = f_iso(Q) + B(Q)/(r_a/a)^2 . At fixed energy the finite positive zero contour is r_a/a = sqrt(-B/f_iso) when B f_iso < 0 . This identity is algebraic, not an independent physical proof of DF admissibility. Results - 3,630 Step 46 input rows; 363 distinct (p, energy-radius) combinations. - Maximum absolute residual of OM affine relation across all original rows: 5.68434e-14. - 139 of 363 combinations have a finite positive zero-contour value of r_a/a (not all within tested anisotropy grid). - 13 energy-radius zero crossings refined by Brent root search using new 192-node quadrature; each independently checked with 96 and 384 nodes. - Maximum absolute 384-node DF at the 192-node roots: 3.46e-9; near steep central transitions, small root errors can magnify the DF residual. Files - [excerpt]

Archived machine-readable result excerpt: {"step46_input_rows": 3630, "affine_identity_max_absolute_error": 5.684341886080802e-14, "affine_identity_median_absolute_error": 1.734723475976807e-17, "energy_p_pairs": 363, "pairs_with_finite_ra_zero": 139, "refined_roots": 13, "root_max_abs_f384": 3.4560662732693745e-09, "root_max_abs_f384_minus_f192": 3.4560726653861935e-09}

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 48 — archived source and numerical evidence

Files (6): README.md, requirements.txt, roots_and_minimum.csv, step47_reference.py, step48.py, summary.json

Archived README synopsis: UDCT v7.14 Step 48 — negative OM distribution-function lobe Exploratory audit, not preregistration. Model: fixed isolated spherical Plummer-generated QUMOND-like acceleration, tracer density rho=(1+(r/a)^p)^(-5/p), OM augmented density rho_OM=(1+r^2/r_a^2)rho. Parameters p=1.5 and r_a/a=0.4. Reproduction Install pip install -r requirements.txt ; run python step48.py . step47_reference.py is bundled for standalone operation. The script computes the DF using the radial integral and a Gauss–Legendre quadrature with a numerical potential difference, searches for the two zero crossings and minimizes the DF between them, and repeats at 96, 192, 384 nodes. Findings See summary.json and roots_and_minimum.csv . The zero crossings in the energy-label radius r/a are 0.2237024398 and 0.7669697146; minimum f=-0.0008538448012 at r/a=0.3804964036. The negative-lobe width in this radius coordinate is 0.5432672748. This is not a physical radial interval containing negative stellar densities: r labels energy, and f<0 denotes failure of the assumed OM DF positivity at those energies. Caveats The 96/192/384 comparisons share the same inversion and gravity assumptions. This does not establish a global positive DF elsewhere, nor rule out other tracer or anisotropy models. No direct Hydrus I individual-star velocity fitting, 3D EFE, or v7.13 24-candidate retest is included. It is not a universal MOND failure theorem. Future work: systematic grid of [excerpt]

Archived machine-readable result excerpt: {"model": "isolated spherical Plummer-generated QUMOND-like law; p=1.5, ra/a=0.4", "left_root_r_over_a": 0.22370243982922897, "right_root_r_over_a": 0.7669697146174498, "negative_interval_width_r_over_a": 0.5432672747882208, "negative_interval_width_log_r": 1.2321305399438338, "minimum_r_over_a": 0.38049640362537734, "minimum_f_192": -0.0008538448012277041, "min_f_96": -0.0008538448011829086, "min_f_384": -0.0008538448012095206, "max_abs_384_minus_192_at_landmarks": 3.967945680355617e-14}

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 49 — archived source and numerical evidence

Files (6): README.md, lobe_map.csv, requirements.txt, step47_reference.py, step49_portable.py, summary.json

Archived README synopsis: UDCT v7.14 — Step 49: OM negative-lobe map (exploratory) This is an exploratory numerical study, not preregistered and not a fit to individual Hydrus I stellar velocities . Model and methods - Plummer-generated spherical gravitational field, QUMOND-like interpolation g = 0.5 gN + sqrt(0.25 gN^2 + a0 gN) . - Tracer density rho(r) = [1+(r/a)^p]^(-5/p) . - Osipkov–Merritt augmented density rho_OM = (1+r^2/ra^2) rho . - Inversion implementation imported from the prior Step 47/48 code, included in this archive as step47_reference.py

. - Parameters: p = 1.49, 1.5, 1.501; ra/a = 0.35, 0.4, 0.45, 0.5, 0.65, 1, infinity. - Search r/a in [0.06,2], 55 log-spaced points. 96-node quadrature for brackets and 192-node quadrature for root/minimum refinement. - lobe_map.csv records bracketed roots, intervals and numerical minima. The r/a variable labels energy via turning radius; it is not a physical orbit-occupancy interval. Main observations 21 parameter pairs, 9 with detected negative intervals in the searched domain. For p=1.5: ra/a roots (r/a) min f --- --- --- 0.35 0.1665216234, 0.8545050504 -0.0020007919203 0.40 0.2237024398, 0.7669697146 -0.0008538448012 0.45 0.3381470120, 0.6206727785 -0.0001583586487 0.50 none found n/a The two roots approach each other as ra/a grows. This does not yet establish the exact double-root critical point: tangent roots can be missed by sign-change searches. A follow-up should jointly solve f=0 and df/d(log [excerpt]

Archived machine-readable result excerpt: {"p_values": [1.49, 1.5, 1.501], "ra_values": [0.35, 0.4, 0.45, 0.5, 0.65, 1.0, "inf"], "r_over_a_scan_range": [0.06, 2.0], "r_grid_points": 55, "model": "isolated spherical Plummer-generated QUMOND-like gravity, tracer [1+(r/a)^p]^(-5/p), OM", "total_parameter_pairs": 21, "pairs_with_negative_interval": 9, "quadrature_n_for_brackets": 96, "quadrature_n_for_roots_minima": 192, "limitations": ["Finite r/a domain; zero roots outside interval untested", "Shared physical inversion kernel; not independent physics check", "Near-tangent zero crossings may escape sign-change scan"]}

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 50 — archived source and numerical evidence

Files (7): README.md, critical_tangencies.csv, p1p5_lobe_collapse.csv, requirements.txt, step47_reference.py, step50.py, summary.json

Archived README synopsis: UDCT v7.14 Step 50 — OM negative-lobe tangency audit Exploratory, not preregistered . Independent of observational stellar-velocity fitting. For the OM augmented density, f(Q;ra)=f_iso(Q)+B(Q)/(ra/a)^2. At a zero, (ra/a)^2=-B/f_iso. A pair of zero crossings merges where this ratio reaches a local maximum with respect to log(r/a). The radius is an energy-label turning radius, not a stellar orbit radius. The script evaluates the original Step 47 quadrature kernel, searches r/a in [0.25,0.75], and checks 96/192/384 point Gauss-Legendre quadratures. A separate scan over [0.15,0.95] tests negative-lobe widths for p=1.5. This is a local finite-domain result, not a proof of global DF nonnegativity or QUMOND failure. Run: python step50.py (requires numpy, scipy, pandas). Outputs: critical_tangencies.csv , p1p5_lobe_collapse.csv , summary.json . No network needed. The bundled step47_reference.py is the computational kernel copied from the previous step. The Step 50 numerical methods and tolerances are fully recorded in step50.py .

Archived machine-readable result excerpt: {"step": 50, "p_values": [1.49, 1.5, 1.501], "critical_tangencies": [{"p": 1.49, "critical_ra_over_a": 0.4627396446066039, "tangent_r_over_a": 0.4763697156418466, "f_at_tangent": 0.0, "df_dlogr_finite_difference": -1.0446593676910476e-09, "critical_ra_n96": 0.462739644606257, "critical_ra_at_same_r_n384": 0.4627396446052533, "delta_ra_n384_minus_n192": -1.350586309456503e-12, "domain": "r/a 0.25 to 0.75; local lobe only"}, {"p": 1.5, "critical_ra_over_a": 0.46646678739601644, "tangent_r_over_a": 0.47409755283222904, "f_at_tangent": 0.0, "df_dlogr_finite_difference": -1.0578452176723818e-09, "critical_ra_n96": 0.4664667873965174, "critical_ra_at_same_r_n384": 0.466466787395183, "delta_ra_n384_minus_n192": -8.33444424586105e-13, "domain": "r/a 0.25 to 0.75; local lobe only"}, {"p": 1.501, "critical_ra_over_a": 0.4668411650904537, "tangent_r_over_a": 0.4738661838352446, "f_at_tangent": 0.0, [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

14. Mathematical checks needed before confirmatory interpretation

1. Re-derive the QUMOND spherical potential with consistent potential zero, boundary and deep-MOND interpolation.

2. Compute central expansion to the first nonzero term at p=1.5 and verify cancellation of all boundary contributions.

3. Implement independent right-sided Abel and conventional relative-potential inversions and compare after a sign-convention transformation.

4. Forward-integrate the recovered DF to reconstruct the input tracer density and quantify residuals over radius.

5. Test full accessible Q, including low-energy and outer-tail limits, rather than only 0.25≤r/a≤0.75.

6. Test alternative tracer laws, source masses and anisotropy families to expose model dependence.

7. Test the Hessian under mesh refinement, coordinate rotations, external-field boundaries and independently differentiated analytic benchmarks.

8. Fit individual stellar velocities and nuisance parameters using a held-out, preregistered observational protocol.

15. Discussion: what the program adds and what it does not

The research value lies in preserving a progressive computational stress test with explicit numerical failures, local parameter boundaries, code and source records. The model-specific critical-exponent argument and OM negative-lobe mapping may help refine future tests. Their novelty relative to existing phase-space consistency literature remains a bibliographic research question rather than a priority claim. This manuscript neither presents a novel complete GQUMOND action nor establishes that the MOND family fails observationally. A negative DF in a chosen model is scientifically useful because it identifies an inadmissible equilibrium construction and motivates more discriminating tests.

16. Conclusions

The two original archives document 50 exploratory stages with a scientifically meaningful separation between 3D Hessian diagnostics, spherical Jeans models and phase-space DF inversions. For the isolated Plummer source/tracer family, the central leading coefficient changes sign on either side of p=1.5 under stated assumptions; exact equality is unresolved by that term. OM scans reveal a reproducible archived local tangency around r_a/a=0.466466787396 at p=1.5 in a specified finite interval, not a global positivity boundary. Hydrus I enters as observational motivation rather than a full stellar likelihood test. A future preregistration should freeze hypotheses, numerical thresholds, datasets and independent-validation protocols before confirmatory inference.

References — external scholarly literature

1. Bekenstein, J. & Milgrom, M. (1984). Does the missing mass problem signal the breakdown of Newtonian gravity? Astrophysical Journal 286, 7–14. https://doi.org/10.1086/162570

2. Milgrom, M. (2010). Quasi-linear formulation of MOND. MNRAS 403, 886–895. https://doi.org/10.1111/j.1365-2966.2009.16184.x

3. Milgrom, M. (2023). Generalizations of quasilinear MOND. Physical Review D 108, 084005. https://doi.org/10.1103/PhysRevD.108.084005

4. Koposov, S. E. et al. (2018). Snake in the Clouds: a new nearby dwarf galaxy in the Magellanic bridge. MNRAS 479, 5343–5361. https://doi.org/10.1093/mnras/sty1772

5. Wolf, J. et al. (2010). Accurate masses for dispersion-supported galaxies. MNRAS 406, 1220–1237. https://doi.org/10.1111/j.1365-2966.2010.16753.x

6. Walker, M. G. & Peñarrubia, J. (2011). A method for measuring (slopes of) the mass profiles of dwarf spheroidal galaxies. Astrophysical Journal 742, 20. https://doi.org/10.1088/0004-637X/742/1/20

7. McGaugh, S. S., Lelli, F. & Schombert, J. M. (2016). Radial acceleration relation in rotationally supported galaxies. Physical Review Letters 117, 201101. https://doi.org/10.1103/PhysRevLett.117.201101

8. Merritt, D. (1985). Spherical stellar systems with spheroidal velocity distributions. MNRAS 214, 25P–28P.

9. Binney, J. & Tremaine, S. (2008). Galactic Dynamics, 2nd ed. Princeton University Press.

References — UDCT research records and provenance

1. Paik, W. S. (2026). UDCT v7.08 GQUMOND prior-art correction. Zenodo DOI: https://doi.org/10.5281/zenodo.23117541

2. Paik, W. S. (2026). UDCT v7.09 preregistration: GQUMOND Example A. Zenodo DOI: https://doi.org/10.5281/zenodo.23120660

3. Paik, W. S. (2026). UDCT v7.09 post-registration. Zenodo DOI: https://doi.org/10.5281/zenodo.23130843

4. Paik, W. S. (2026). UDCT v7.10 preregistration sensitivity. Zenodo DOI: https://doi.org/10.5281/zenodo.23200693

5. Paik, W. S. (2026). UDCT v7.10 post-registration. Zenodo DOI: https://doi.org/10.5281/zenodo.23233837

6. Paik, W. S. (2026). UDCT v7.11 preregistration. Zenodo DOI: https://doi.org/10.5281/zenodo.23244522

7. Paik, W. S. (2026). UDCT v7.11 post-registration. Zenodo DOI: https://doi.org/10.5281/zenodo.23263099

8. Paik, W. S. (2026). UDCT v7.12 preregistration. Zenodo DOI: https://doi.org/10.5281/zenodo.23280972

9. Paik, W. S. (2026). UDCT v7.12 post-registration. Zenodo DOI: https://doi.org/10.5281/zenodo.23296444

10. Paik, W. S. (2026). UDCT v7.13 preregistration. Zenodo DOI: https://doi.org/10.5281/zenodo.23342971

11. Paik, W. S. (2026). UDCT v7.13 post-registration. Zenodo DOI: https://doi.org/10.5281/zenodo.23539127

12. Paik, W. S. (2026). UDCT v7.13 post-registration II. Zenodo DOI: https://doi.org/10.5281/zenodo.23559995

Source provenance note: prior UDCT DOI values were supplied in the author’s publication record; the present compilation has not independently verified every historical DOI landing page. Original v7.14 numerical artifacts are preserved in the companion ZIP.

UDCT v7.14 Exploratory Findings: The p=1.5 Boundary in QUMOND

A model-specific asymptotic threshold and finite-energy Osipkov–Merritt tangency audit

Won Shik Paik | Independent Researcher | Auckland, New Zealand

Status: Exploratory research note (not preregistered; not a confirmatory observational test). 28 October 2026.

Reserved Zenodo DOI: 10.5281/zenodo.23695048 (not registered until publication).

Code repository: https://github.com/wspaik5/UDCT_v7.14_Exploratory_p1.5_Boundary

Abstract

This technical note consolidates the archived UDCT v7.14 exploratory Steps 01–50 and six subsequent numerical audits. In a specified isolated spherical simple-QUMOND Plummer-field construction, the tracer family rho_p(r)=[1+(r/a)^p]^(-5/p) has a central potential scaling Psi proportional to r^(3/2), implying a leading tracer-density expansion proportional to Psi^(2p/3). The coefficient of its second derivative changes sign across p=3/2 and vanishes at p=3/2. This asymptotic observation must not be conflated with a universal QUMOND threshold or an exact finite-energy distribution-function zero. Separate Osipkov–Merritt (OM) inversions show a model-specific local zero-DF tangency near ra/a=0.4664667874 at p=1.5. A six-point p scan, finite-domain sign checks, quadrature convergence, and two numerical optimizers support a smooth local critical curve, not a discontinuity at p=1.5. The calculations share a common physical inversion kernel and do not establish global DF positivity, dynamical stability, a 3D external-field result, or direct Hydrus I stellar likelihood agreement.

1. Scientific question and evidential scope

The motivating question is whether the central asymptotic exponent p=1.5 and the finite-energy OM anisotropy tangency represent one physical transition. The archived results instead require a two-boundary interpretation: an asymptotic coefficient cancellation and a continuously varying local tangency condition. The present document is a synthesis of existing exploratory artifacts, not a newly preregistered prediction. No post hoc numerical criterion is relabeled as confirmatory.

2. Source archives and provenance

The user-provided primary archives are “7.14 Hessian 1~25.zip” (early exploratory calculations), “7.14 Hessian 26~50.zip” (25 nested step packages), and “7.14 Exploratory audit 1~6.zip” (six nested audit packages). Audit 06 is separately available as an executable nine-file archive. A SHA-256 inventory is provided in the companion provenance manifest. A nested archive name is evidence of archival presence, not independent verification of every embedded result.

3. Defined model and normalization

The Step 40/Audit 01 model specifies G=4.30091e-3 pc (km/s)^2/Msun, M=32730 Msun, a=53.3 pc, a0=1.2e-10 m/s^2; a Plummer gravitating mass with Newtonian acceleration gN=G M r/(r^2+a^2)^(3/2); and the isolated spherical simple-QUMOND acceleration g=gN/2+sqrt(gN^2/4+a0 gN). The tracer density is rho_p=[1+(r/a)^p]^(-5/p), normalized to rho(0)=1. Potential and inversion conventions follow the archived scripts; numerical DF magnitudes are therefore convention- and normalization-dependent.

4. Derivation of the central asymptotic exponent

At r much smaller than a, gN is proportional to r. In the deep-acceleration central limit, the leading square-root term makes g proportional to r^(1/2). Integration from the center gives Psi(r)=Phi(r)-Phi(0) proportional to r^(3/2). Meanwhile rho_p(r)=1-(5/p)(r/a)^p+O(r^(2p)). Eliminating r gives rho_p(Psi)=1-C Psi^alpha+... with alpha=2p/3 and C positive under the stated conventions. Hence the leading second derivative is -C alpha(alpha-1) Psi^(alpha-2). Its coefficient vanishes at alpha=1, i.e. p=1.5, and the sign of this leading term reverses on crossing that exponent. At p=1.5 the omitted next-order terms determine the local behavior. Therefore the leading-term cancellation is not by itself a theorem that the full DF vanishes at all energies or that a global physical transition occurs.

5. Finite-energy OM inversion and local tangency

For the archived OM inversion, the numerical implementation uses an affine decomposition f(Q,ra)=f_iso(Q)+B(Q)/(ra/a)^2. When f_iso is positive and B negative, the local zero-DF condition gives (ra/a)^2=-B/f_iso. Maximizing this local critical-radius expression over a finite energy-label interval yields a candidate tangency. The energy-label r/a in these audits is an inversion-energy coordinate and should not be misread as an observed stellar orbital radius. This procedure neither certifies positivity over the full energy continuum nor proves that every distribution function is dynamically realizable.

6. Quantitative local critical curve

p	Critical ra/a (192 nodes)	Energy-label r/a
1.490	0.462739644607	0.476369776
1.495	0.464599468922	0.475243086
1.499	0.466092716172	0.474328250
1.500	0.466466787396	0.474097647
1.501	0.466841165090	0.473866280
1.505	0.468341767989	0.472933131

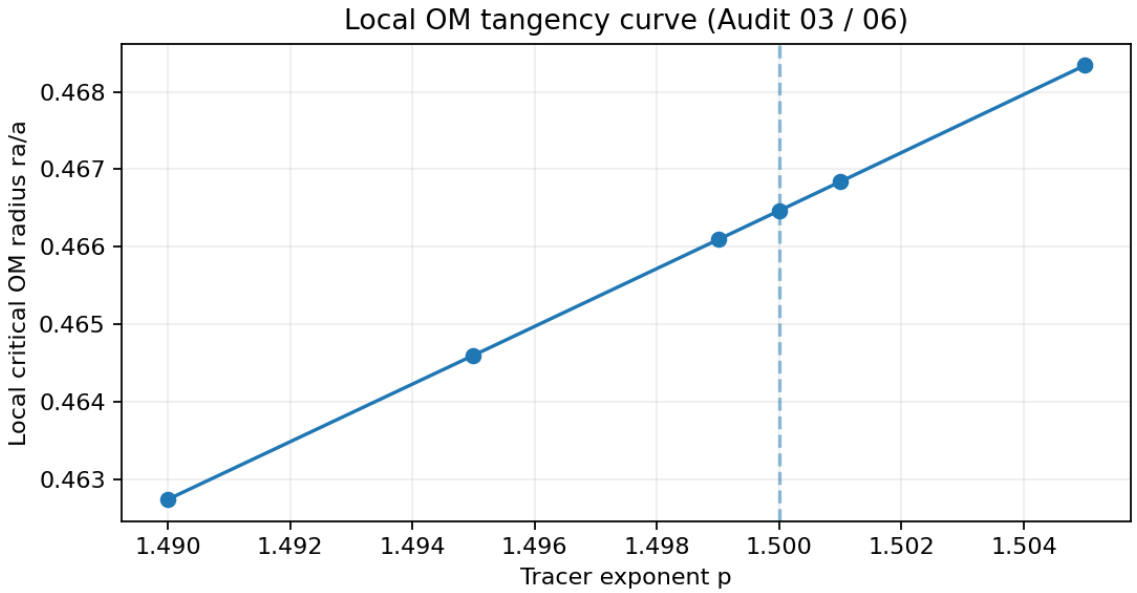


Figure 1. Local OM critical-radius curve from the archived Audit 03 input re-tabulated in Audit 06; this is not a global boundary.

7. Numerical cross-checks and convergence

Audit 04 evaluated 3 tracer exponents by 4 anisotropy settings by 18 energy labels (216 DF samples). At ra/a=0.45, seven sampled negative DF values were reported for each p=1.499, 1.500 and 1.501. At the p=1.5 reference radius ra/a=0.466466787396, p=1.501 yielded three negative samples whereas p=1.499 and 1.500 yielded none under the stated threshold f<-1e-10. These counts are finite-grid diagnostics, not global sign classifications.

Audit 05 fixed p=1.501 and ra/a=0.466466787396. The reported local minima were -3.2713868467e-6, -3.2713868390e-6 and -3.2713868336e-6 for 96, 192 and 384 Gauss–Legendre nodes, respectively. At 192 nodes the adjusted local zero-tangency radius was ra/a approximately 0.466841165090. This supports numerical persistence of a small negative lobe in that finite interval, but not exact error certification.

Audit 06 compared bounded minimization and golden-section search for p=1.499, 1.500 and 1.501; the maximum reported discrepancy in critical ra/a was 5.6e-17, reflecting agreement of optimization on the same numerical function. Central finite differences gave slope 0.37422446 and curvature 0.30646998 at h=0.001, versus slope 0.37422991 and curvature 0.30648475 at h=0.005. Their agreement supports local smoothness within the sampled model and numerical procedure, not analyticity or universal criticality.

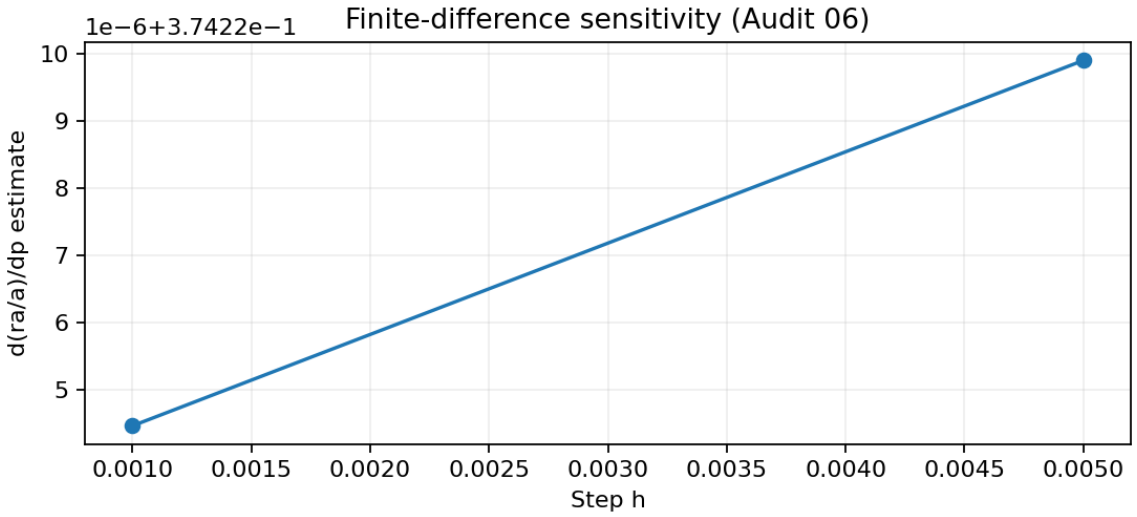


Figure 2. Finite-difference slope sensitivity at p=1.5; only two step sizes were archived.

8. Audit-by-audit evidence ledger

Audit	Question	What is supported
01	Asymptotics and independent quadrature	Leading exponent cancellation and numerical agreement; not a new inversion theorem
02	Finite energy scan	No resolved significant negative on the sampled reference tangency grid; not continuum positivity
03	Variation with p	Local critical ra/a changes continuously over sampled p values
04	Crossing p=1.5	Finite-grid sign differences distinguish asymptotic and OM conditions
05	Quadrature convergence	Persistent local negative lobe for p=1.501 at fixed reference anisotropy
06	Optimizer and derivative checks	Algorithm agreement and stable local finite-difference slope/curvature

9. Interpretation and falsifiability

The evidence supports two distinct mathematical features in the specified spherical model. The p=1.5 coefficient cancellation is an asymptotic statement; the OM local tangency is an energy- and anisotropy-dependent numerical

condition. The observed smooth movement of the latter across p=1.5 is incompatible with treating p=1.5 as an automatically singular finite-energy OM radius. A later preregistration should freeze independent inversion implementations, global energy domains, numerical tolerances, and model variations before evaluating new outcomes. In particular, a positive DF on a finite sample must not be described as a proof of phase-space admissibility over all energies.

10. Hydrus I and observational boundaries

Hydrus I provides observational motivation and model-inspired input scales in portions of the broader UDCT program. The archived v7.13 and v7.14 internal Jeans/Hessian and phase-space stress tests are not equivalent to a direct likelihood fit to the complete individual-star line-of-sight velocity data. No such fit is demonstrated in this note. The 3D external-field problem, membership uncertainty, observational systematics, and full dynamical stability require separately defined tests.

11. Limitations and independence

All six audits share physical assumptions with the archived Step 47 field and inversion kernel. Different quadrature resolutions and optimization algorithms test numerical stability but are not independent derivations of the physical model. The scans use finite intervals and grids; extremely small differences between algorithms may exceed realistic physical accuracy. No certified interval arithmetic, full continuum proof, general nonspherical GQUMOND calculation, or parameter-free observational discrimination is claimed. The model includes fixed mass, scale and acceleration inputs; parameter-free should not be inferred from the absence of a fit in the current exercise.

12. Roadmap and preregistration separation

The present note should be published as exploratory findings first. A distinct v7.14 preregistration may subsequently state novel falsifiable questions, freeze model classes and datasets, preregister thresholds and negative controls, and require independent code paths. Binary systems, Bullet Cluster lensing and cross-scale invariance remain long-term future directions only; no universality is claimed from a shared numerical exponent.

13. Future Directions: Near-Term Tests and Long-Term Extensions

This section distinguishes prospective tests motivated by the present exploratory audit from broader research ambitions. None of the activities below is represented as a completed result or as evidence of a universal p = 1.5 threshold.

12.1 Near-term: v7.14 preregistration and confirmatory tests

• Freeze the exact QUMOND interpolation, tracer family, boundary conditions, nondimensionalization, Osipkov–Merritt inversion convention, energy domain, and numerical tolerances before new runs. Explicitly separate the central asymptotic coefficient cancellation at p = 1.5 from the finite-energy local tangency curve.

• Preregister new p values and anisotropy radii not used to locate the exploratory boundary; specify a priori success/failure thresholds, convergence criteria, negative controls, and an independent computational implementation. Retain null and contradictory outcomes.

• Test sensitivity to central regularization, quadrature order, optimization brackets, finite-energy sampling, and alternative tracer profiles. Assess global distribution-function nonnegativity over a prespecified energy domain rather than extrapolating from local minima.

• Extend the consistency checks toward stationary Jeans admissibility, three-dimensional Hessian diagnostics, and phase-space distribution-function reconstruction, distinguishing mathematical consistency from dynamical stability.

• Only after the model-level tests are frozen should observational confrontation be planned: use documented Hydrus I inputs and, in a separate stage, stellar line-of-sight velocities with selection functions, measurement uncertainties, membership modeling, and likelihood-based comparison. The present study does not perform that direct fit.

12.2 Longer-term: independent astrophysical and cross-scale tests

• Binary-star and wide-binary systems: construct a separate dynamical and external-field model and test it against suitable astrometric and kinematic data, with conventional gravity and existing MOND formulations as controls.

• Bullet Cluster and other merging galaxy clusters: develop a time-dependent, three-dimensional treatment including baryonic mass maps, gas dynamics, lensing observables, and merger geometry. A static or local p = 1.5 result alone cannot address cluster lensing offsets.

• Cross-scale and multiscale invariance: test whether any dimensionless boundary structure survives changes of mass profile, size, environment, acceleration regime, and observable. Reappearance of a similar exponent in one additional system would not by itself demonstrate universality.

• If the model-level consistency tests succeed, explore two-way field–matter back-reaction and alternative dynamical sectors; evaluate conservation laws, well-posedness, stability, and independent observational predictions before advancing a new gravity claim.

Publication sequence: exploratory findings (this note) → separately frozen v7.14 preregistration → post-registration execution and audit → later extensions only if justified by the results.

14. Conclusions

In the specified isolated spherical QUMOND tracer model, p=1.5 marks a cancellation of the leading central asymptotic second-derivative coefficient. The archived OM inversion has a separate finite-energy local tangency at ra/a approximately 0.4664667874 for p=1.5, with a smoothly varying critical curve around that exponent. Audits 01–06 improve traceability and local numerical convergence checks, but do not establish a universal QUMOND threshold, global DF positivity, stability, or observational confirmation. The scientifically defensible result is a bounded, reproducible model-specific distinction that motivates genuinely prospective preregistered tests.

References — External Scholarly Literature

1. Bekenstein, J. & Milgrom, M. (1984). Does the missing mass problem signal the breakdown of Newtonian gravity? Astrophysical Journal 286, 7–14. https://doi.org/10.1086/162570

2. Milgrom, M. (2010). Quasi-linear formulation of MOND. MNRAS 403, 886–895. https://doi.org/10.1111/j.1365-2966.2009.16184.x

3. Milgrom, M. (2023). Generalizations of quasilinear MOND. Physical Review D 108, 084005. https://doi.org/10.1103/PhysRevD.108.084005

4. Koposov, S. E. et al. (2018). Snake in the Clouds: a new nearby dwarf galaxy in the Magellanic bridge. MNRAS 479, 5343–5361. https://doi.org/10.1093/mnras/sty1772

5. Wolf, J. et al. (2010). Accurate masses for dispersion-supported galaxies. MNRAS 406, 1220–1237. https://doi.org/10.1111/j.1365-2966.2010.16753.x

6. Walker, M. G. & Peñarrubia, J. (2011). A method for measuring (slopes of) the mass profiles of dwarf spheroidal galaxies. Astrophysical Journal 742, 20. https://doi.org/10.1088/0004-637X/742/1/20

7. McGaugh, S., Lelli, F. & Schombert, J. (2016). Radial acceleration relation in rotationally supported galaxies. Physical Review Letters 117, 201101. https://doi.org/10.1103/PhysRevLett.117.201101

8. Merritt, D. (1985). Spherical stellar systems with spheroidal velocity distributions. MNRAS 214, 25P–28P.

9. Binney, J. & Tremaine, S. (2008). Galactic Dynamics, 2nd ed. Princeton University Press.

References — UDCT Research Records and Provenance

The prior UDCT version sequence (v7.08–v7.13) is cited as the author's research provenance, not as independent peer-reviewed confirmation. Specific historical DOI strings should be verified against published Zenodo records before final release. Primary materials actually used here are the user-provided v7.14 Steps 01–50 archive pair and six audit ZIPs; their hashes and file listings are included in the companion manifest.

1. Paik, W. S. (2026). UDCT v7.14 exploratory Steps 01–50. User-provided archives: "7.14 Hessian 1~25.zip" and "7.14 Hessian 26~50.zip".

2. Paik, W. S. (2026). UDCT v7.14 Exploratory Audits 01–06. User-provided archive: "7.14 Exploratory audit 1~6.zip"; executable audit subarchives.

3. Paik, W. S. (2026). UDCT v7.14 Exploratory Findings: The p=1.5 Boundary in QUMOND. Reserved DOI 10.5281/zenodo.23695048; https://github.com/wspaik5/UDCT_v7.14_Exploratory_p1.5_Boundary (publication pending).

4. Paik, W. S. (2026). UDCT v7.08–v7.13 preregistration, post-registration and GQUMOND prior-art correction research records. Verify individual DOI records against Zenodo prior to public citation.

Source provenance and reproducibility statement

The companion package includes exact SHA-256 hashes of the three source ZIPs, the six audit ZIPs, and the Audit 06 tables, together with the unmodified Audit 06 executable archive. The figures in this note were generated from the archived Audit 06 CSV values. No new independent full-physics rerun is claimed for this synthesis. Readers can reproduce the original local calculations from the archived audit scripts and their declared dependencies.

Appendices A–F: Detailed Audit Record

These appendices are newly typeset explanatory material appended after the six original pages. The original PDF pages, including their text, tables, figures, conclusions, two reference sections and original pagination, are retained without rewriting. Appendix numerical statements are transcribed from the six archived audit packages; no fresh full-physics rerun is claimed.

Notation: p = tracer-shape exponent; a = model scale; r_a = Osipkov–Merritt anisotropy radius; r/a in the inversion calculations is an energy label, not an observed orbital radius; DF = distribution function. Numerical precision in printed tables is not a certified error bound.

Audit archive manifest

Audit	Original ZIP	SHA-256 (prefix)
01	UDCT_v7.14_Exploratory_Audit_01.zip	ca0343666bb7cf502374...
02	UDCT_v7.14_Exploratory_Audit_02.zip	0dd6e73ea6ef93dd7134...
03	UDCT_v7.14_Exploratory_Audit_03.zip	1d9bbc7ac9938812b7dd...
04	UDCT_v7.14_Exploratory_Audit_04.zip	c99c748c500fbce3072f...
05	UDCT_v7.14_Exploratory_Audit_05.zip	e86c5c45c9d5e8c794f1...
06	UDCT_v7_14_Exploratory_Audit_06.zip	71178608a1d634acfd12...

A. Audit 01 — Asymptotic and independent-quadrature verification

A.1 Objective and inputs

Audit 01 compares the Step 40 central inverse-series argument with a symbolic derivative check and selected Step 50 finite-energy Osipkov–Merritt inversions with an alternative adaptive quadrature. The archive also includes unchanged original Step 40 and Step 50 scripts and results. Both comparisons are exploratory, and the quadrature calculation retains the source gravitational and inversion kernels.

A.2 Asymptotic argument

For the Plummer gravitating mass, g_N(r)=GMr/(r²+a²)^(3/2), so g_N is proportional to r at small r. The isolated simple-QUMOND acceleration g=g_N/2+sqrt(g_N²/4+a_0 g_N) is proportional to r^(1/2) at leading order. The central potential difference therefore scales as Psi proportional to r^(3/2). The tracer rho_p(r)=[1+(r/a)^p]^(-5/p) has rho_p=1-(5/p)(r/a)^p+...; consequently rho_p(Psi)=1-C Psi^(2p/3)+..., C>0. The leading second derivative is -C alpha(alpha-1)Psi^(alpha-2), alpha=2p/3, and its coefficient vanishes at p=1.5.

This is cancellation of one asymptotic term. Higher-order terms matter exactly at p=1.5; the full finite-energy DF need not vanish. This distinction is the primary mathematical caution motivating the later audits.

A.3 Adaptive quadrature and evidence

Diagnostic	Archived value
Integration	SciPy adaptive quad with t=sqrt(log(r/r0)) substitution
Evaluated samples	15
Negative samples	1

Diagnostic	Archived value
Integration warnings	0

Selected original adaptive-quadrature table rows (see companion CSV for all columns and full precision):

p	ra_over_a	energy_label_r_over_a	df
1.5	0.46	0.474097552832229	-5.757231477049456
1.5	0.466466787396	0.01	0.0185117053081338
1.5	0.466466787396	0.2	0.0015030157810879
1.5	0.466466787396	0.35	0.0001692605929428
1.5	0.466466787396	0.43	1.59098149014528e-
1.5	0.466466787396	0.47	1.1909331261691536
1.5	0.466466787396	0.474097552832229	1.5274142900158996

A.4 Reproduction and limits

Run symbolic_step40_audit.py and independent_quad_audit.py after installing the included requirements. The adaptive results cover 15 selected points only. They do not certify a continuum sign, and agreement between integration methods cannot rule out shared errors in the model or inversion formula.

B. Audit 02 — Extended finite-energy sign scan

B.1 Question and numerical protocol

At the p=1.5 local tangency identified in the exploratory Step 50 calculation, does a finite energy-label scan resolve a negative DF? Audit 02 fixes p=1.5 and r_a/a=0.466466787396, evaluates 61 energy-label points on r/a in [0.01,30], and performs bounded local minimization on [0.36,0.60]. The included step47_reference.py is the common physical kernel.

B.2 Archived numerical outcomes

Quantity	Result
Tracer exponent p	1.5
OM radius r_a/a	0.466466787396
Energy-label domain	[0.01, 30.0]
Grid samples	61
Strictly negative grid values	1
Minimum sampled DF	-1.429771859502e-16
Minimum grid energy label	0.474097552832
Locally optimized DF	-1.422916815544e-16
Locally optimized energy label	0.474097563301
Max absolute N96 vs N192 difference	2.300317922255e-12

B.3 Interpretation

The negative minimum is approximately 1.4e-16 in the archive's DF normalization, i.e. numerically indistinguishable from a zero tangency at the reported precision without a certified error interval. The single negative grid value is not evidence of a robust negative lobe. Conversely, a finite scan cannot prove that the DF is nonnegative at all energies.

B.4 Reproduction and limitations

Install requirements and run audit02.py. It regenerates finite_domain_scan.csv and audit02_results.json. The grid is finite, the minimization is local, and the inversion kernel is shared with the earlier step. No individual-star Hydrus I likelihood or external-field simulation is performed.

C. Audit 03 — Critical anisotropy curve near p=1.5

C.1 Purpose and local critical-radius construction

Audit 03 asks whether the local zero-DF tangency r_a/a is stationary or discontinuous at p=1.5. It uses the affine Osipkov–Merritt decomposition $f(Q,r_a)=f_{iso}(Q)+B(Q)/(r_a/a)^2$. Where $f_{iso}>0$ and $B<0$, the candidate local critical radius is $\sqrt{-B/f_{iso}}$. The algorithm maximizes this candidate over the restricted energy-label interval r/a in [0.25,0.75].

C.2 Six-point archived curve

p	Critical r_a/a (N192 at N96 optimum)	Energy label r/a
1.490	0.462739644607	0.476369776
1.495	0.464599468922	0.475243086
1.499	0.466092716172	0.474328250
1.500	0.466466787396	0.474097647
1.501	0.466841165090	0.473866280
1.505	0.468341767989	0.472933131

C.3 Quadrature-order comparison

The archived 96-node optimization is followed by a 192-node evaluation at the same energy label; this is not a full re-optimization at 192 nodes. Absolute N192–N96 critical-radius differences in the six rows range from roughly 5e-14 to 1e-12. These small differences indicate numerical stability of this specific calculation but not physical accuracy to twelve decimal places.

C.4 Interpretation and limits

The sampled local critical radius rises from about 0.462739645 at p=1.490 to 0.468341768 at p=1.505. The local tangency energy label shifts downward over the same range. This is evidence for a smoothly varying finite-energy local boundary in the specified model, distinct from the p=1.5 central asymptotic coefficient cancellation. It does not establish a global positivity boundary, analyticity, or applicability to other tracer families.

Reproduce with audit03.py, which writes critical_vs_p.csv and results.json. The search bracket, quadrature order and physical kernel are fixed as archived.

D. Audit 04 — Parameter-grid sign stress test

D.1 Protocol

Audit 04 evaluates 3 tracer exponents p={1.499,1.500,1.501}, 4 OM settings r_a/a={0.45,0.466466787396,0.50,infinity}, and 18 energy-label values r/a between 0.01 and 30. This is 216 DF evaluations. The code uses 64-point Gauss–Legendre quadrature and the exact affine decomposition from the shared Step 47 kernel. A sample is counted negative only when f<-1e-10.

D.2 All twelve archived parameter combinations

p	r_a/a	Negative / 18	Minimum sampled DF
1.499	0.45	7	-1.5432351e-04
1.499	0.466466787396	0	4.9534452e-07
1.499	0.5	0	4.3116419e-07
1.499	inf	0	2.5942292e-10
1.500	0.45	7	-1.5818197e-04
1.500	0.466466787396	0	6.6530982e-15
1.500	0.5	0	4.3122411e-07
1.500	inf	0	2.5945275e-10
1.501	0.45	7	-1.6204600e-04
1.501	0.466466787396	3	-3.2653149e-06
1.501	0.5	0	4.3128381e-07
1.501	inf	0	2.5948246e-10

D.3 Interpretation

At r_a/a=0.45, seven sampled values are negative for each of the three p values. At the p=1.5 reference tangency r_a/a=0.466466787396, no values cross the negative threshold for p=1.499 or 1.500, whereas three do for p=1.501. At r_a/a=0.50 and in the isotropic limit, none of the 18 samples cross the threshold for any of the three p values. These are finite-grid counts, not continuum sign classifications.

D.4 Reproduction and limitations

Run audit04.py to regenerate finite_domain_sign_scan.csv and summary.json. The threshold -1e-10 is a numerical classification convention, not a physical transition criterion. Near-zero results require higher quadrature order and local minimization, addressed in Audit 05. No independent model implementation or observational likelihood is involved.

E. Audit 05 — High-resolution negative-lobe convergence

E.1 Fixed model and search

Audit 05 holds p=1.501 and r_a/a=0.466466787396 fixed, then minimizes the DF over the finite energy-label interval r/a in [0.25,0.75]. It compares Gauss–Legendre quadrature with N=96, 192 and 384 nodes. It also performs a separate local zero-tangency search for r_a/a in [0.45,0.50] at 192 nodes.

E.2 Archived convergence results

Nodes	Minimum DF	Minimum energy label r/a
96	-3.2713868467321e-06	0.473167428686
192	-3.2713868390085e-06	0.473167428914
384	-3.2713868336334e-06	0.473167421101

The minimum stays near -3.27138684e-6 across the three quadrature orders, substantially larger in magnitude than the near-zero rounding-scale residual of Audit 02. The 192-node local zero-tangency result at p=1.501 is r_a/a=0.466841165090 at energy label r/a=0.473866183640.

E.3 Scientific significance

The convergent negative lobe at p=1.501 and the p=1.5 reference anisotropy radius is a more robust local numerical sign result than a single nearly-zero grid point. It supports movement of the local critical anisotropy radius with p. It is not evidence of a singular universal transition at p=1.5: the tested r_a/a was deliberately held at the p=1.5 local tangency.

E.4 Remaining uncertainties

Quadrature convergence and optimizer convergence are distinct. The reported digits do not provide certified total errors. The local search interval excludes other energy regions, and a shared inversion kernel could introduce common systematic error. Run audit05.py to regenerate minima_convergence.csv and summary.json.

F. Audit 06 — Optimizer agreement and local derivatives

F.1 Purpose and methods

Audit 06 cross-checks the local tangency maximization using bounded scalar optimization and golden-section search on the shared numerical function. The optimizer comparison uses energy labels r/a in [0.43,0.52] at p=1.499, 1.500 and 1.501. The finite-difference derivative estimates at p=1.5 use the archived Audit 03 curve at two step sizes; they are not analytic derivatives.

F.2 Optimizer comparison

p	Bounded r_a/a	Golden r_a/a	Abs. difference
1.499	0.466092716172	0.466092716172	0.000e+00
1.500	0.466466787397	0.466466787397	5.551e-17
1.501	0.466841165091	0.466841165091	0.000e+00

F.3 Finite-difference slope and curvature

h	Estimated slope	Estimated curvature
0.001	0.374224460	0.306469978
0.005	0.374229907	0.306484746

The largest reported optimizer discrepancy in r_a/a is 5.55e-17, reflecting agreement in floating-point evaluation of the same objective. The slope estimates near 0.37422 and curvature near 0.30648 are stable at the two tested finite-difference spacings within their reported variation. Two spacings alone cannot certify smoothness or analyticity beyond the sampled region.

F.4 Reproduction, provenance and interpretation

Run audit06.py with step47_reference.py and the archived Audit 03 critical_vs_p.csv. Outputs are optimizer_comparison.csv, curve_derivatives.csv, results.json and run.log. This is an algorithmic numerical cross-check, not an independent physical inversion. The local tangency curve appears smooth in the tested neighborhood, so the central p=1.5 cancellation and finite-energy OM tangency must remain conceptually separate.

Appendix G — Reproducibility map and audit boundaries

This appendix adds an explicit source map without modifying the six-page main article. All original six ZIP packages are preserved separately. The accompanying appendix ZIP contains these six original audit archives without recompression of their internal members, a SHA-256 inventory, and this complete merged PDF.

Audit	Primary script	Primary output
01	symbolic_step40_audit.py; independent_quad_audit.py	adaptive_quad_results.csv; adaptive_quad_summary.json
02	audit02.py	finite_domain_scan.csv; audit02_results.json
03	audit03.py	critical_vs_p.csv; results.json
04	audit04.py	finite_domain_sign_scan.csv; summary.json
05	audit05.py	minima_convergence.csv; summary.json
06	audit06.py	optimizer_comparison.csv; curve_derivatives.csv; results.json

The six audit archives provide evidence of exploratory numerical checks under the specified spherical QUMOND and Osipkov–Merritt assumptions. They do not establish global phase-space admissibility, dynamical stability, a new gravity law, or a likelihood fit to individual Hydrus I stellar velocities. The proposed v7.14 preregistration must separately freeze new tests and decision thresholds before confirmatory execution.